भारतीय सूचना प्रौद्योगिकी संस्थान नागपुर

Indian Institute of Information Technology Nagpur

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Regression Analysis

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Outline

1. Introduction
2. Linear Regression Model
3. Least Square Method
4. Performance Evaluation Measures
5. Types of Regression Analysis
6. Industry Use Cases

Given a Dataset

Training Eg. or Instance	Input Features						Output
	X1	**X2**	**X3**	**.**	**.**	**Xn**	**Y**
I1	a1	a2	a3	.	.	an	Y1
I2	b1	b2	b3	.	.	bn	Y2
I3	c1	c2	c3	.	.	cn	Y3
.							
.							
.							
Im	P1	P2	P3	.	.	Pn	Ym
Test Input	**Z1**	**Z2**	**Z3**	**.**	**.**	**Zn**	**??**

Classification or Regression ??

Model has to predict it.

Supervised Learning-

For each input x, the desired output y is given. Here y is the label.

Pre-classified training example-

Given: x, y pairs.

for any unseen value of x, determine the best label y.

Classification - When y is discrete

Eg: Masked or not

Red or Black

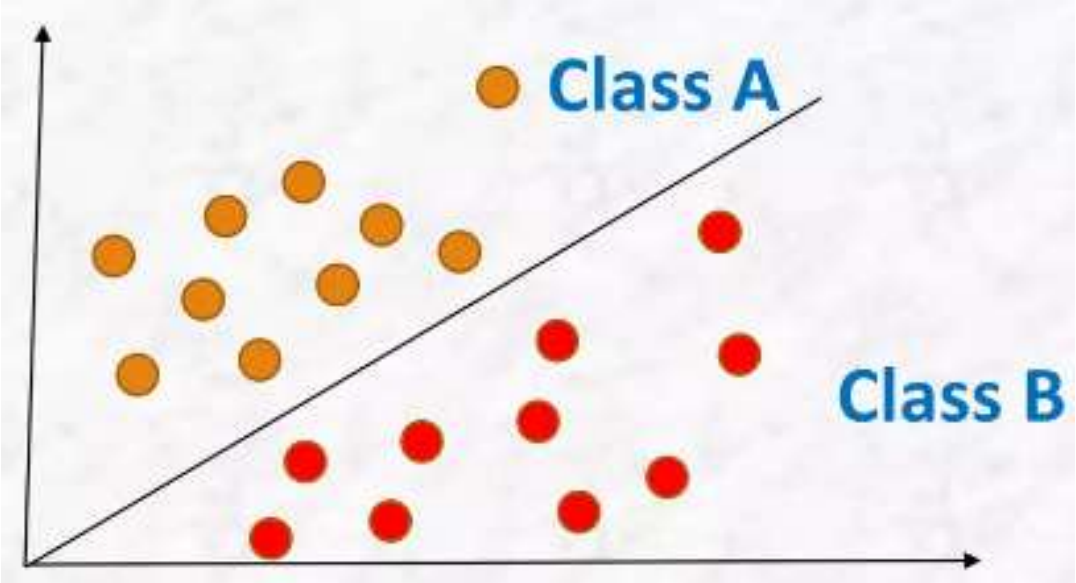
Cat, Dog, Horse

Regression - When y is continuous.

Eg: carpet area, location - predict the price of the house.

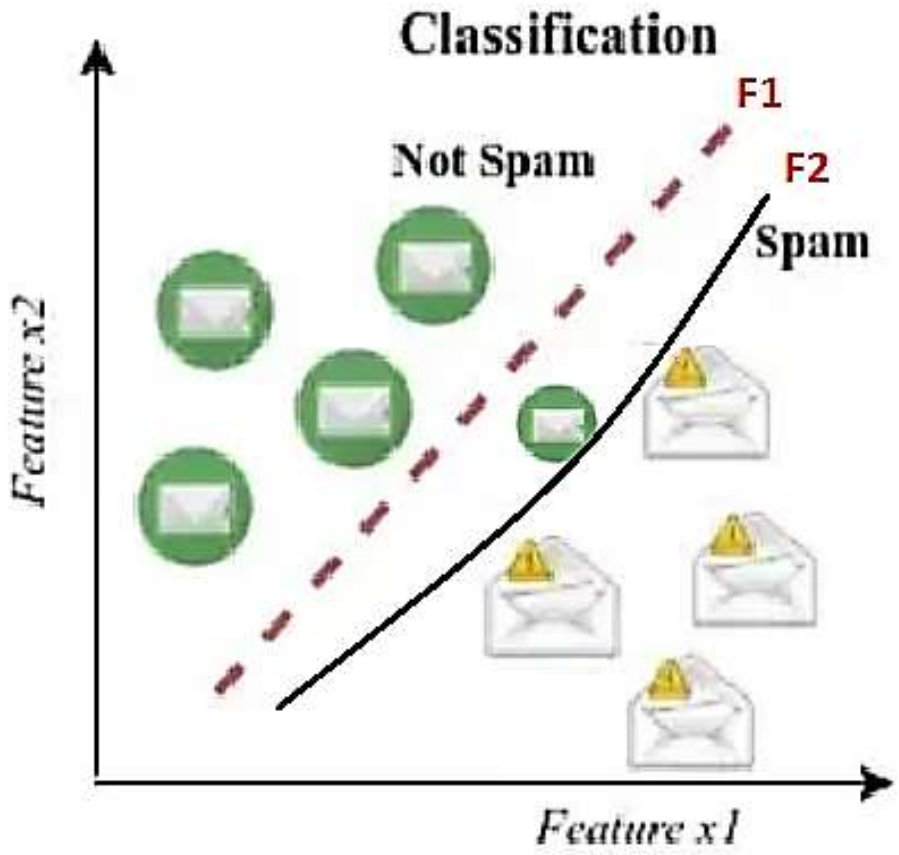
Classification

- Classification is a process of categorizing a given set of data into classes.
- It can be performed on both structured or unstructured data.
- The process starts with predicting the class of given data points. The classes are often referred to as target, label or categories.



Classification Example:-

You are given with the collection of emails, determine the spam or non-spam email from it.



Here the task is to learn the function which can classify and predict if the email is spam or non-spam.
The line F2 is better than F1.

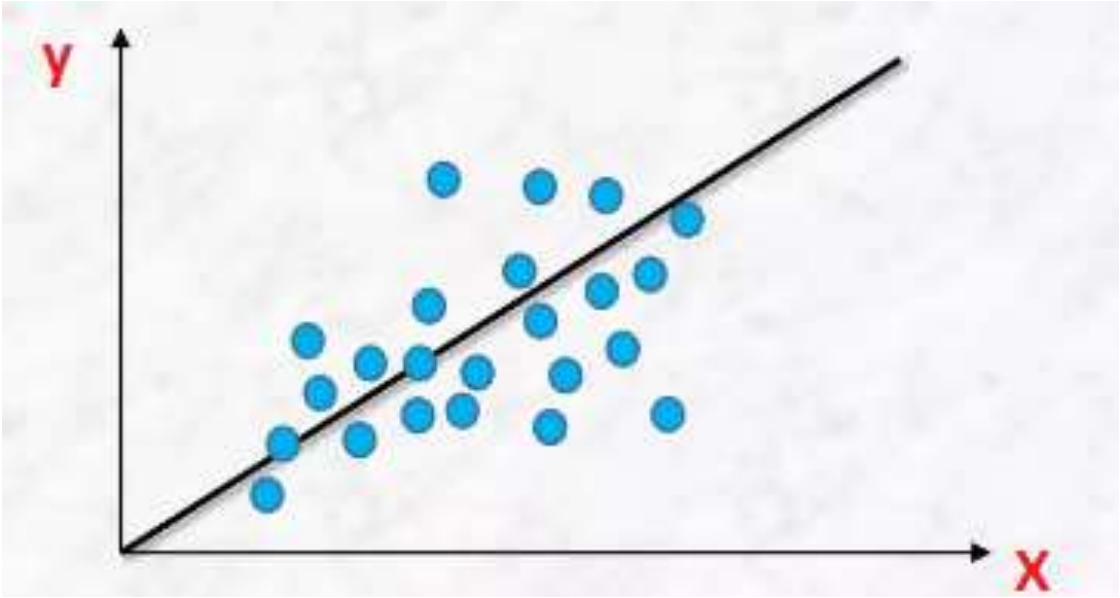
Classification Algorithms

- Decision Tree
- K-NN
- SVM
- ANN

Regression

- A technique for determining the statistical relationship between two or more variables where a change in a dependent variable is associated with, and depends on, a change in one or more independent variables.

- A regression problem is used when the output variable is a real or continuous value, such as "Salary" or "weight".



Regression Example –

- Sales of a product can be predicted by using the relationship between sales volume and amount of advertising.
- The performance of a student can be predicted by using the relationship between hours of study and score in the exam.
- The size of a child's vocabulary can be predicted by using the relationship between the vocabulary size, the child's age and the parents' educational input.

Regression Techniques

- Linear Regression
- Multiple Regression

Regression Example –

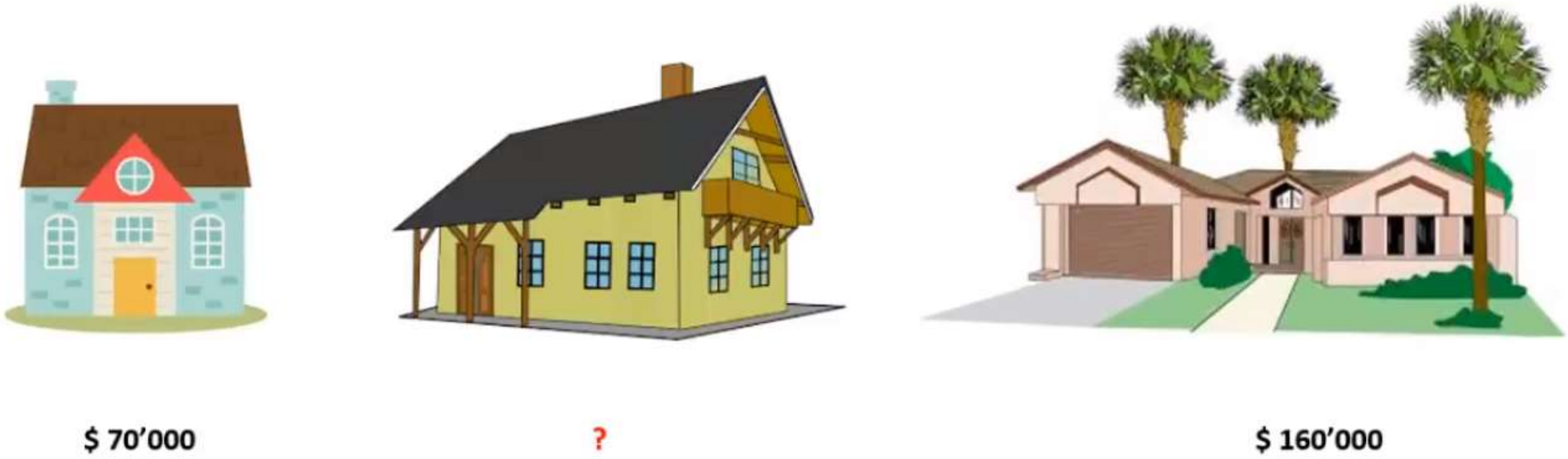
Estimate the price of the house from the given data.

$ 70'000

$ 160'000

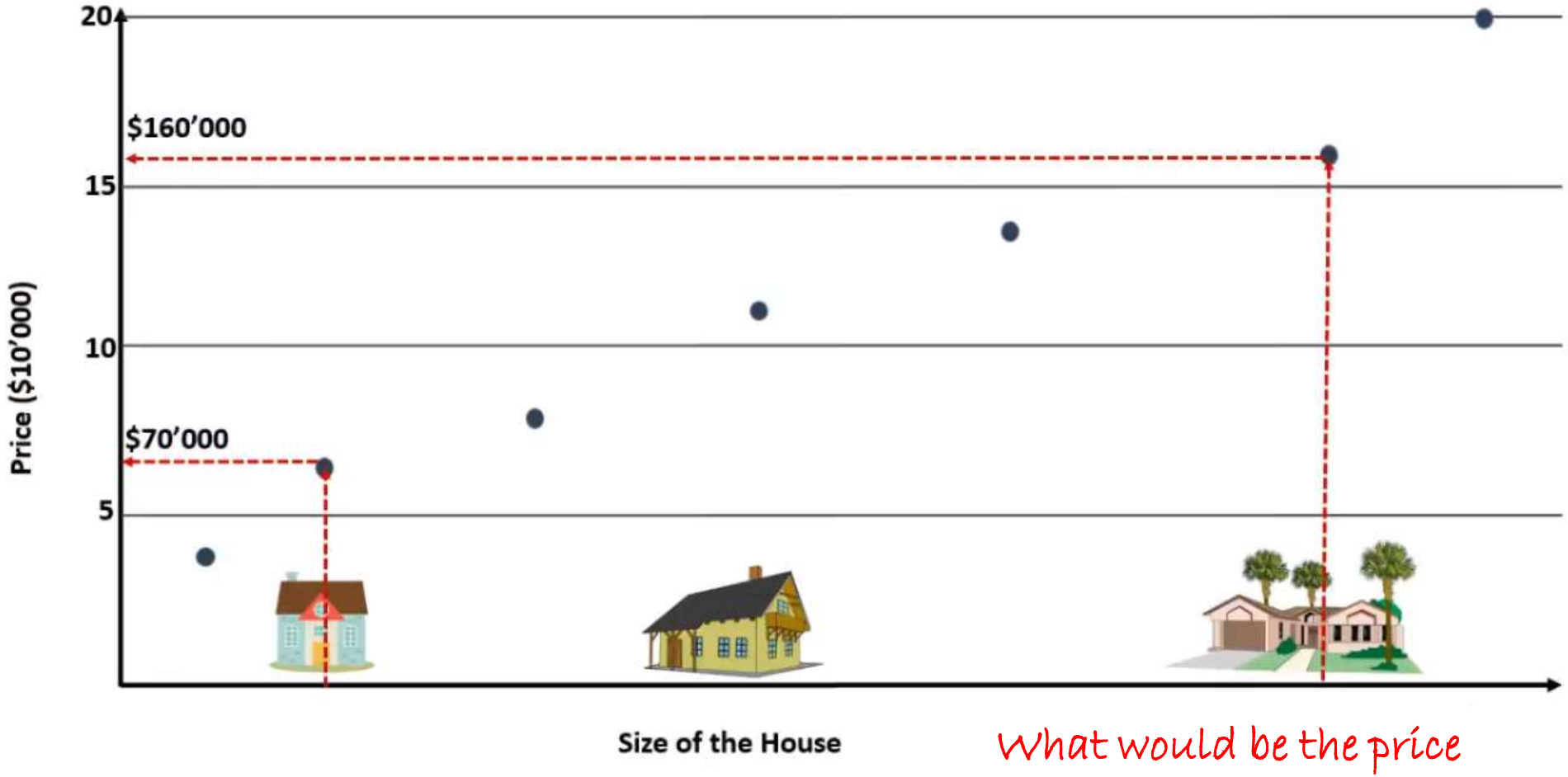
Regression Example –

Estimate the price of the house from the given data.



Regression Example –

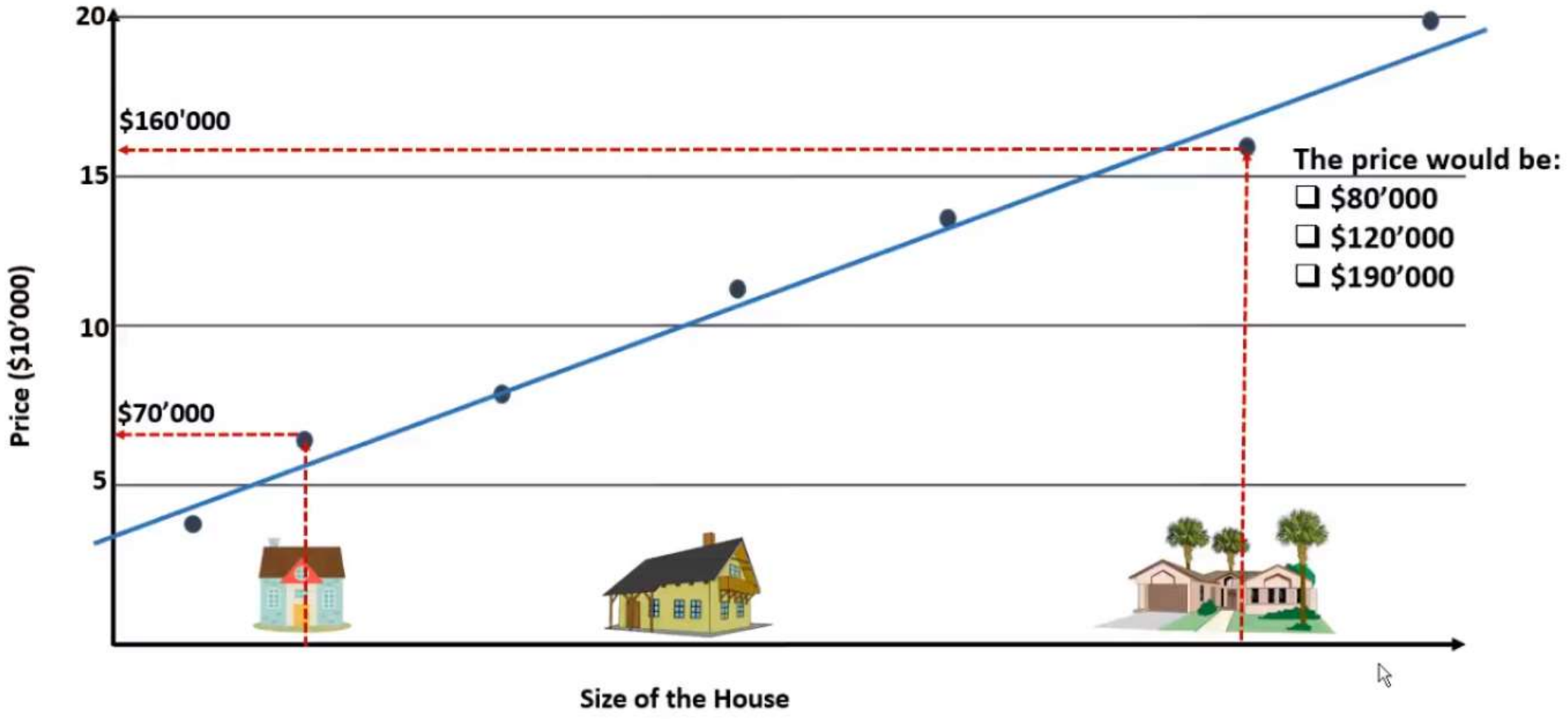
Estimate the price of the house from the given data.



What would be the price of the medium size house

Regression Example –

Estimate the price of the house from the given data.



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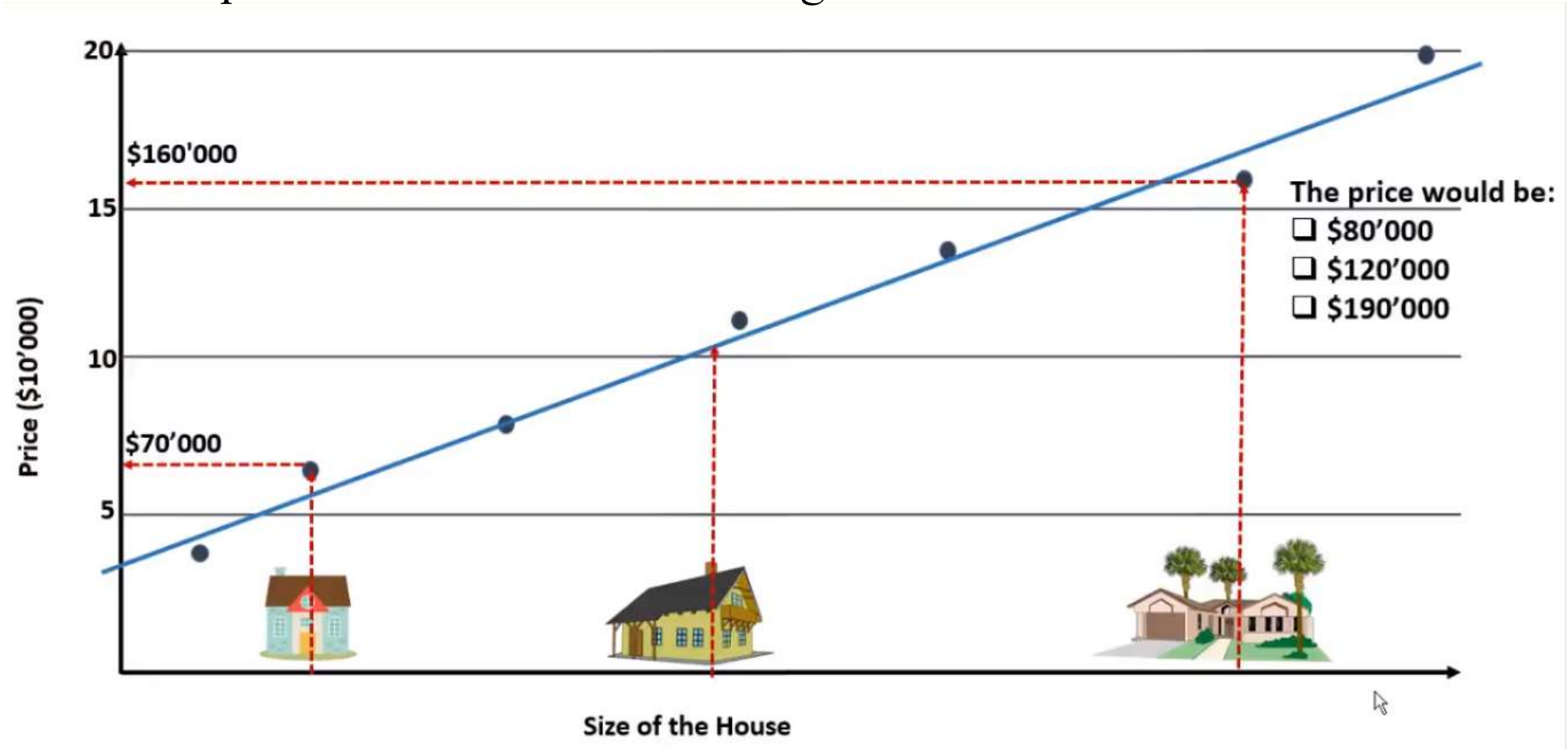

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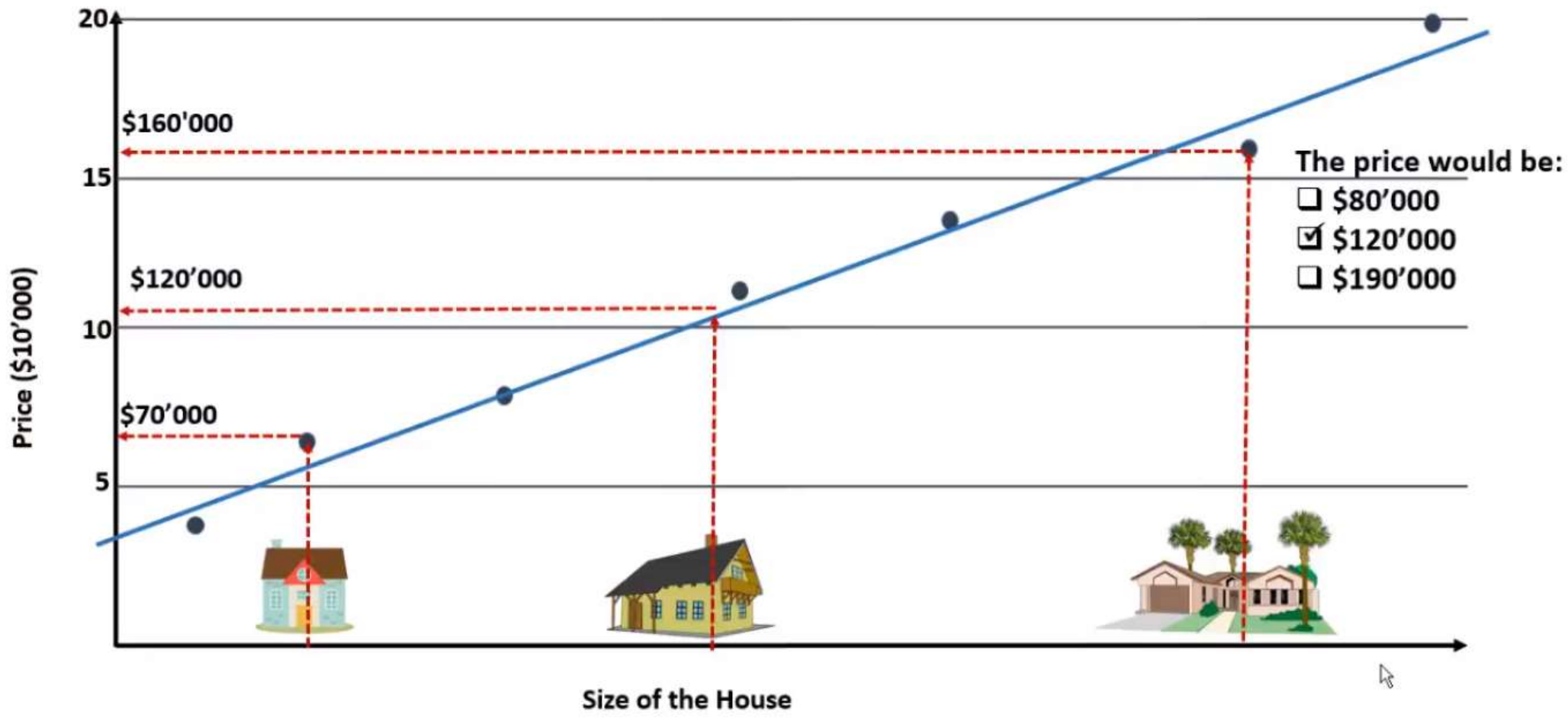
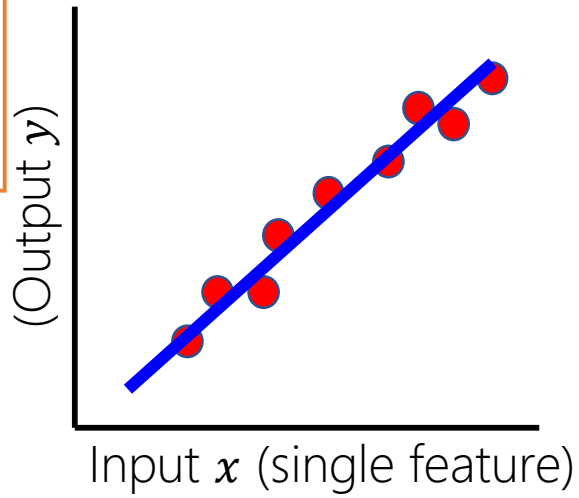


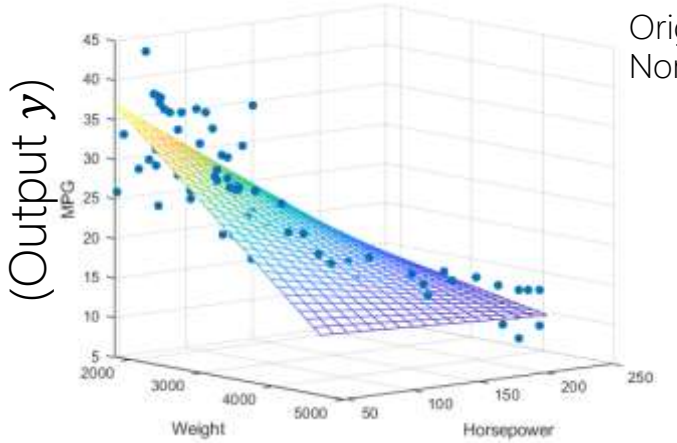
Linear Regression: Pictorially

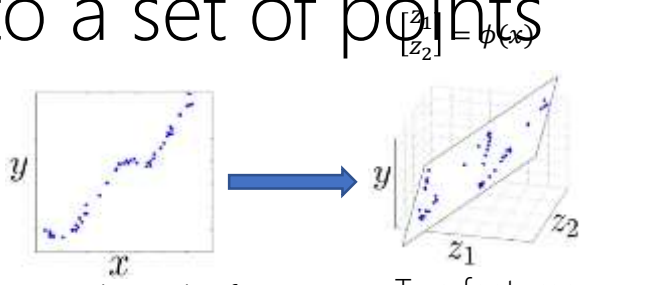
- Linear regression is like fitting a line or (hyper)plane to a set of points

What if a line/plane doesn't model the input-output relationship very well, e.g., if their relationship is better modeled by a nonlinear curve or curved surface?

Do linear models become useless in such cases?







Original (single) feature
Nonlinear curve needed

Two features
Can fit a plane (linear)

No. We can even fit a curve using a linear model after suitably transforming the inputs

$$y \approx \boldsymbol{w}^{\top}\boldsymbol{\phi}(\boldsymbol{x})$$

The transformation $\boldsymbol{\phi}(.)$ can be predefined or learned (e.g., using kernel methods or a deep neural network based feature extractor). More on this later

- The line/plane must also predict outputs the unseen (test) inputs well

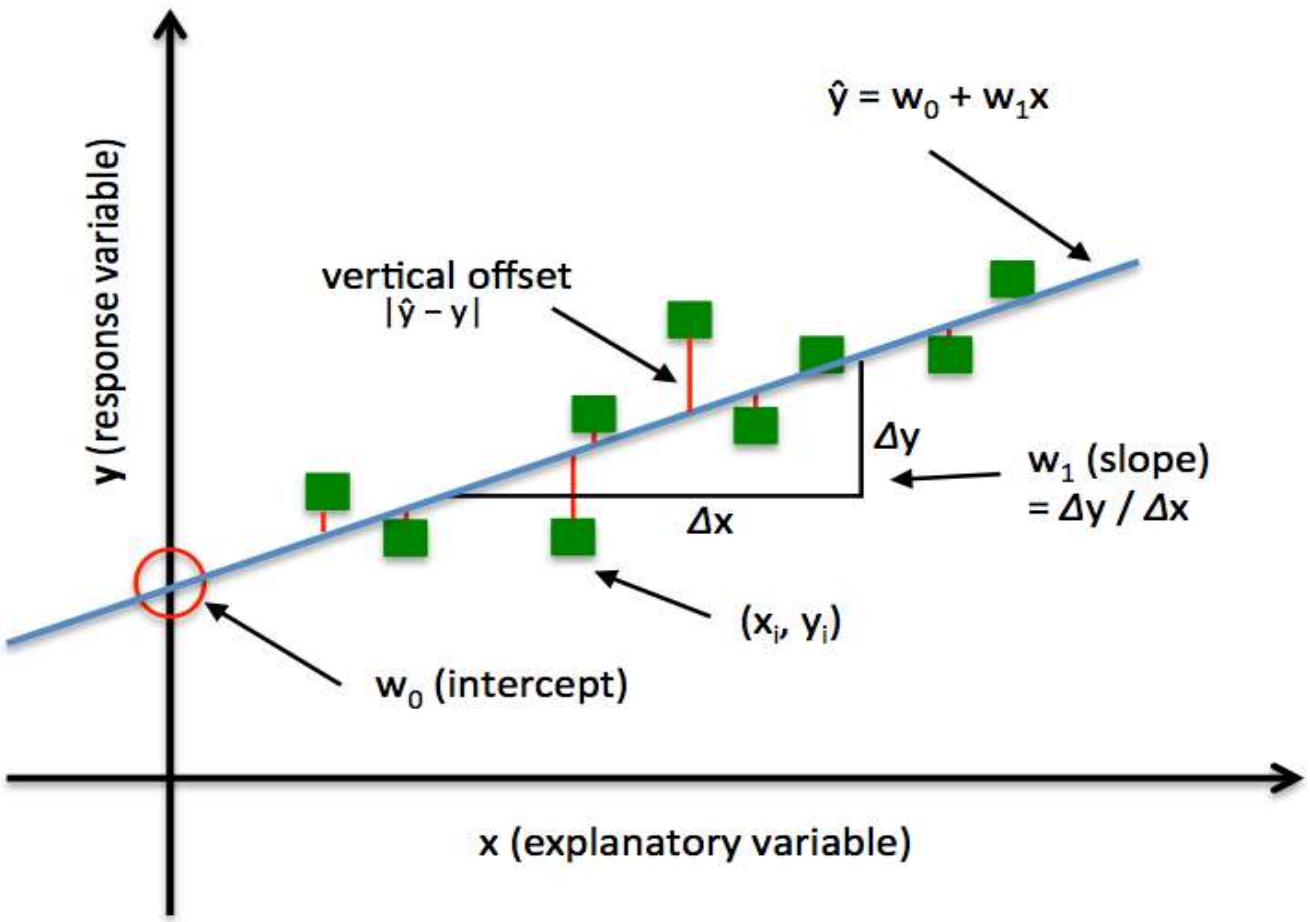
Simplest Possible Linear Regression Model

- This is the base model for <u>all</u> statistical machine learning
- x is a one feature data variable
- y is the value we are trying to predict
- The regression model is

$$y = w_0 + w_1 x + \varepsilon$$

Two parameters to estimate – the slope of the line w_1 and the y-intercept w_0

- ε is the unexplained, random, or error component.



Dependent and Independent Variable

- Independent variables are considered as an input to a system and may take on different values freely.
- Dependent variables are those values that change as a consequence of changes in other values in the system.
- Independent variable is also called as predictor or explanatory variable and is denoted by X
- Dependent variable is also called as response variable and is denoted by Y.

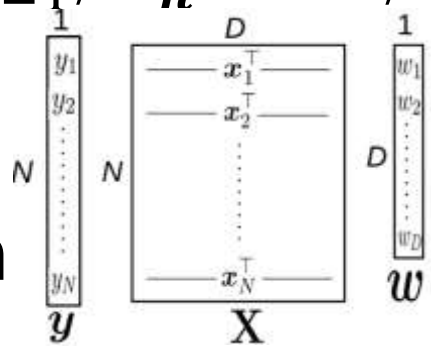
Linear Regression

- The Simplest mathematical relationship between two variables x and y is a linear relationship
- In a cause and an effect relationship, the independent variable is cause, and the dependent variable is the effect.
- **Least squares linear regression** is a method for predicting the value of a dependent variable Y, based on the value of an independent variable X.

$$\hat{y}=mx+b$$

More generally

- Given: Training data with N input-output pairs $\{(\boldsymbol{x}_n, y_n)\}_{n=1}^{N}$, $\boldsymbol{x}_n \in \mathbb{R}^D$, $y_n \in \mathbb{R}$



- Goal: Learn a model to predict the output for new test in

- Assume the function that approximates the I/O re
linear model

$$y_n \approx f(\boldsymbol{x}_n) = \boldsymbol{w}^\top \boldsymbol{x}_n \quad (n = 1,2,\ldots,N)$$

Can also write all of them compactly using matrix-vector notation as $\boldsymbol{y} \approx \boldsymbol{X}\boldsymbol{w}$

Goal of learning is to find the $\boldsymbol{w}$ that minimizes this loss + does well on test data

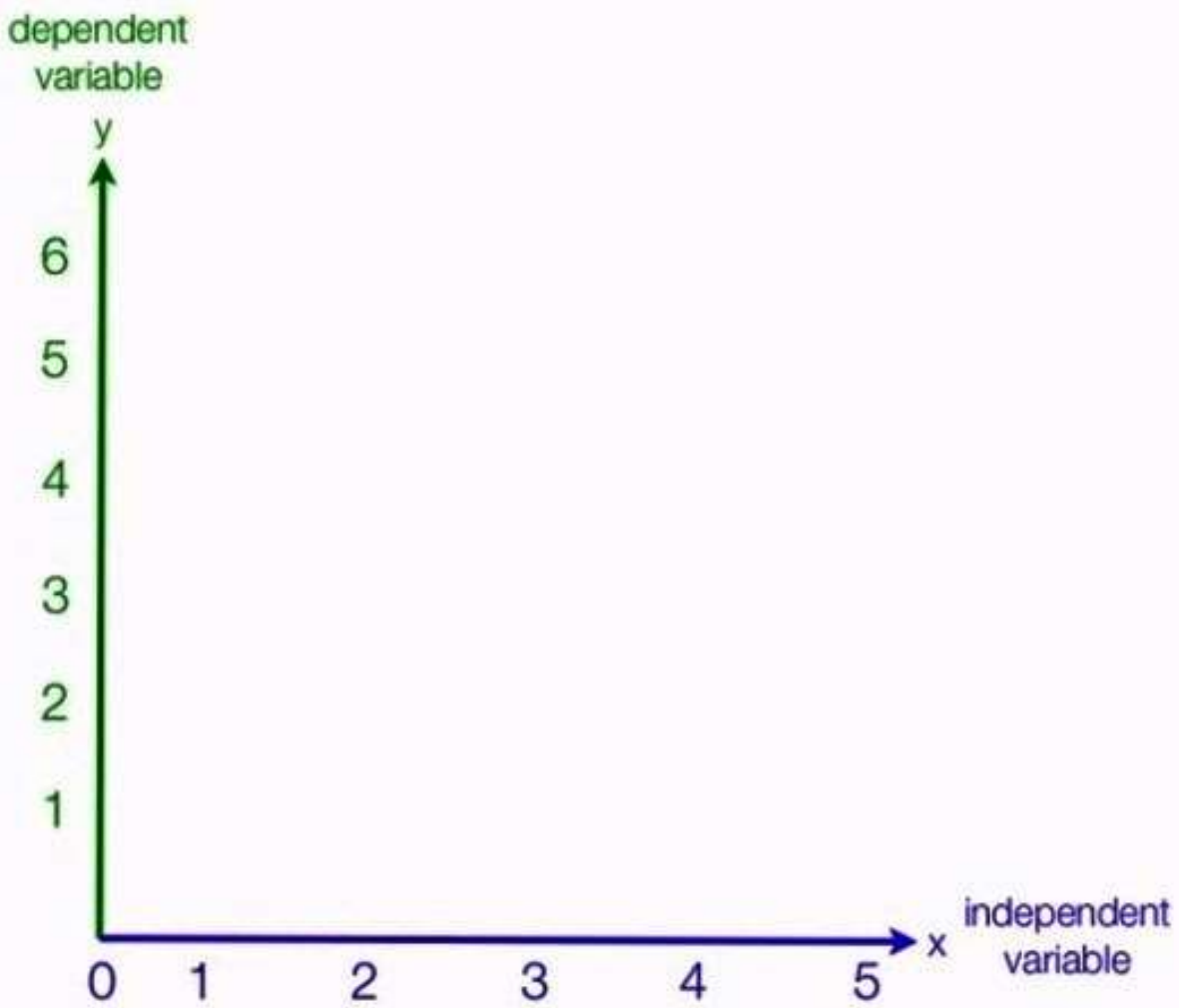
$$L(\boldsymbol{w}) = \sum_{n=1}^{N} \ell(y_n, \boldsymbol{w}^\top \boldsymbol{x}_n)$$

otal error or loss of this mo

Unlike models like KNN and DT, here we have an <u>explicit problem-specific objective</u> (loss function) that we wish to optimize for

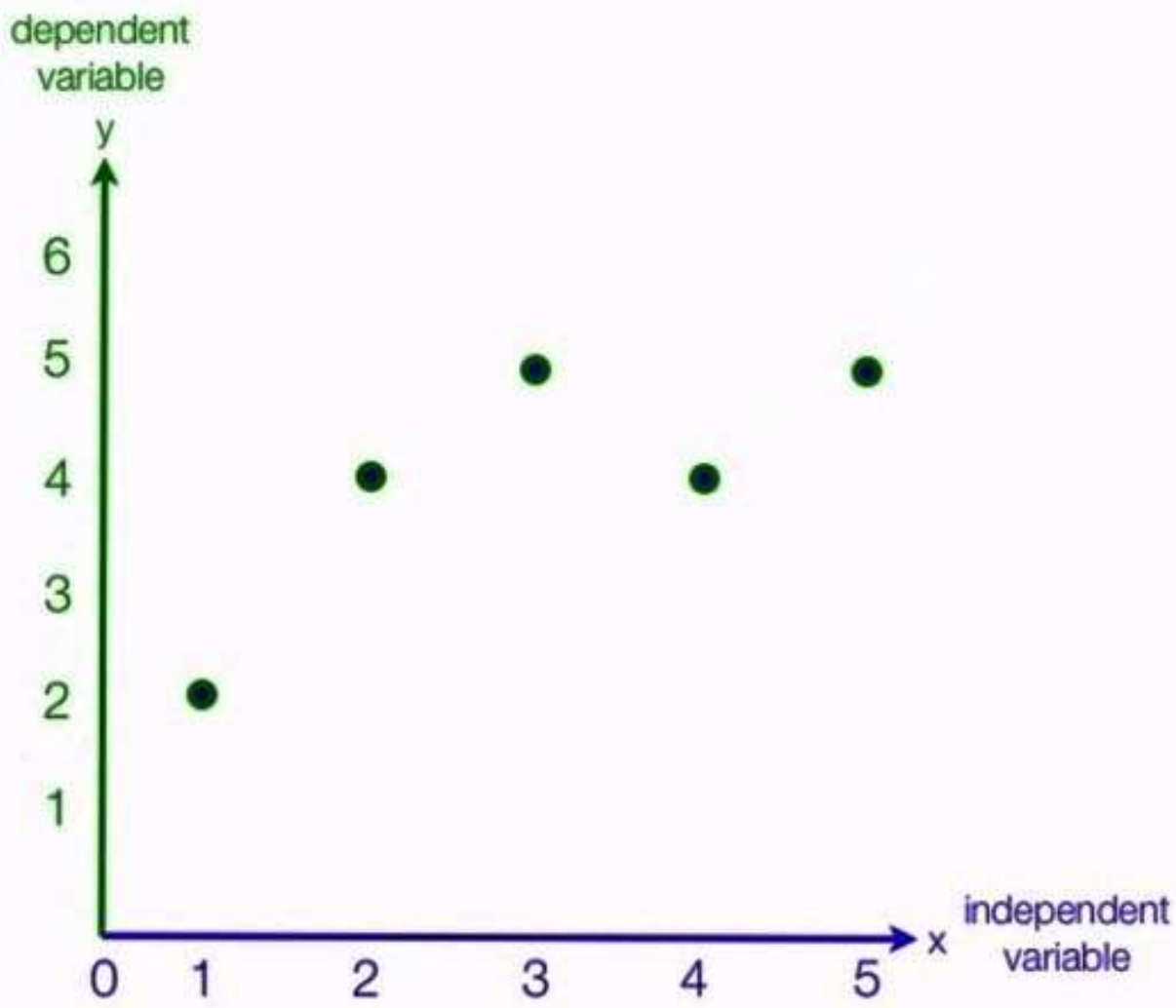
$\ell(y_n, \boldsymbol{w}^\top \boldsymbol{x}_n)$ measures the prediction error or "loss" or "deviation" of the model on a single training input $(\boldsymbol{x}_n, y_n)$

Linear Regression using Least Square Method



independent variable	dependent variable
x	y
1	2
2	4
3	5
4	4
5	5

Linear Regression using Least Square Method



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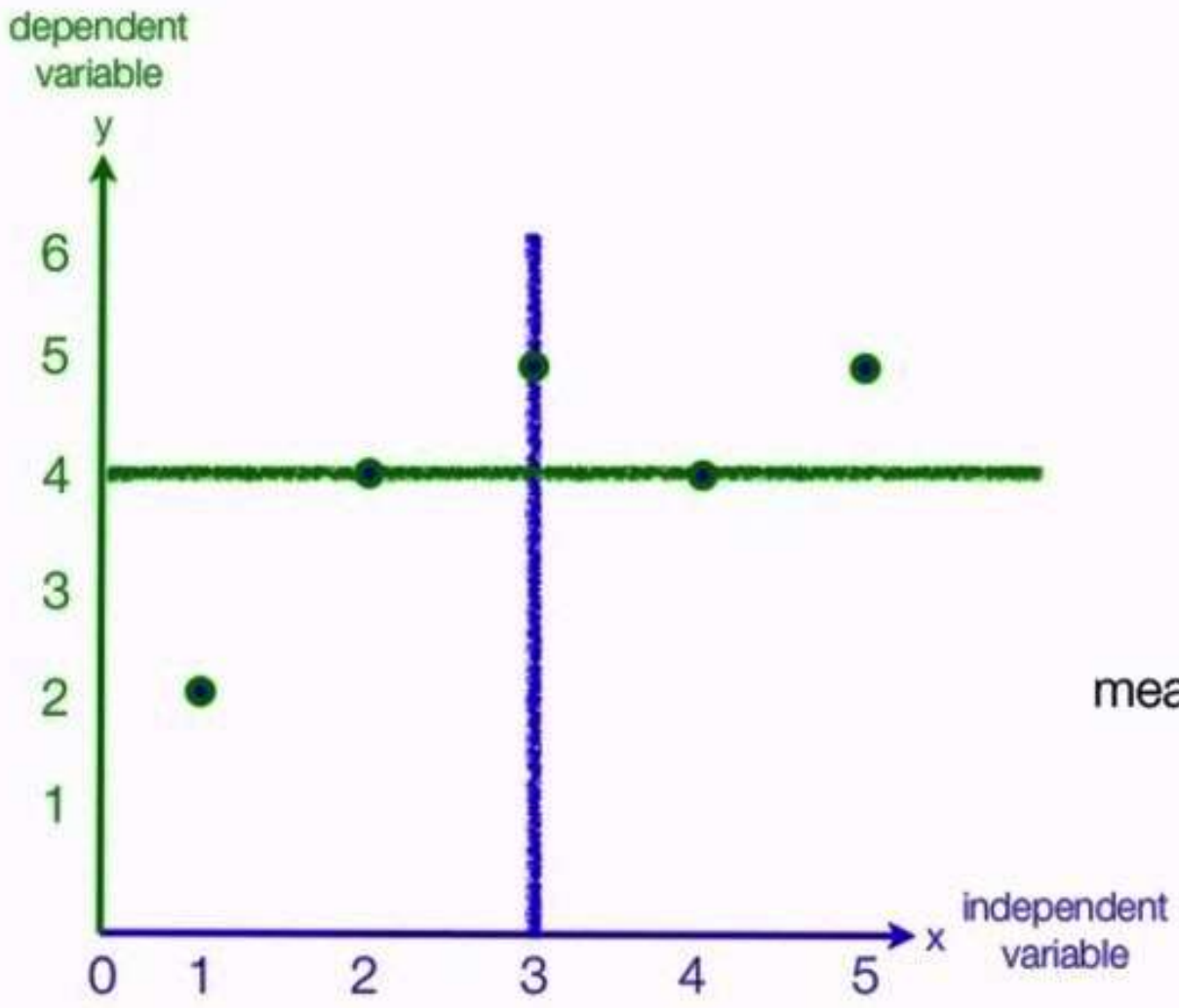

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mean	3	4

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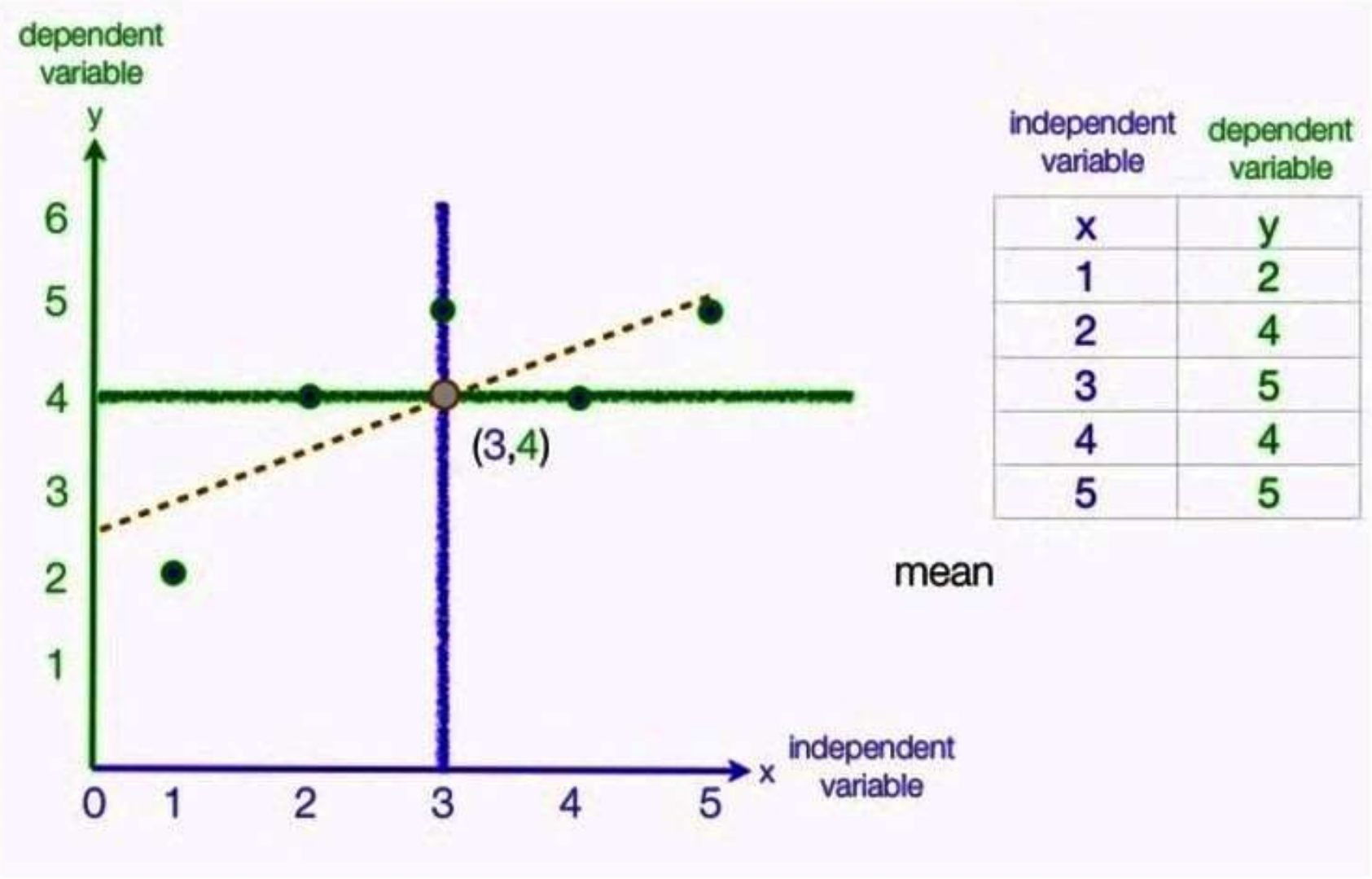

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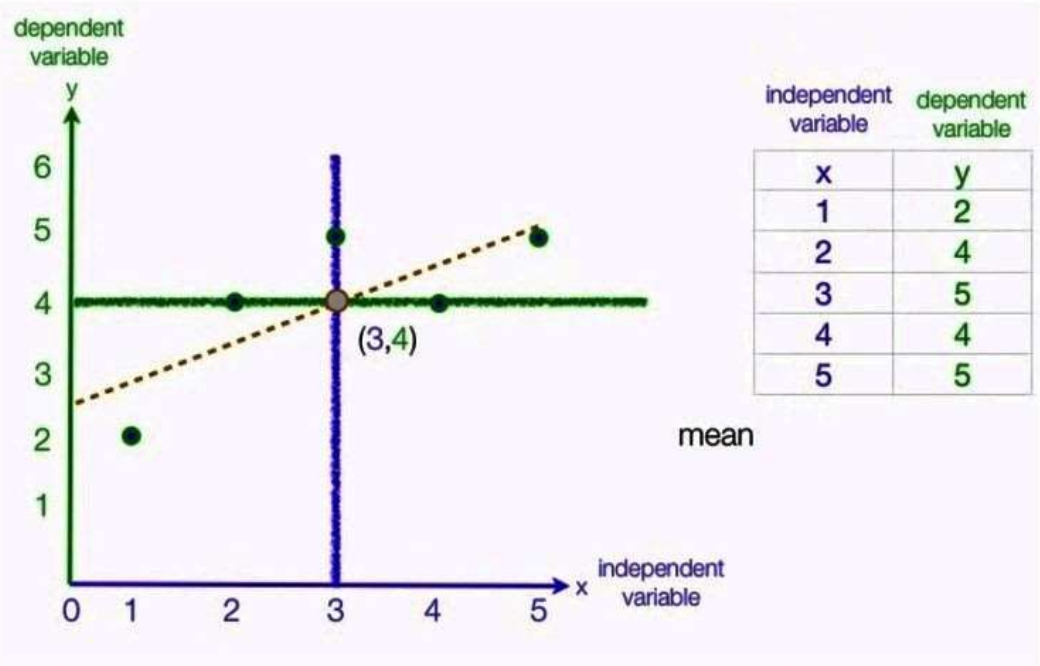

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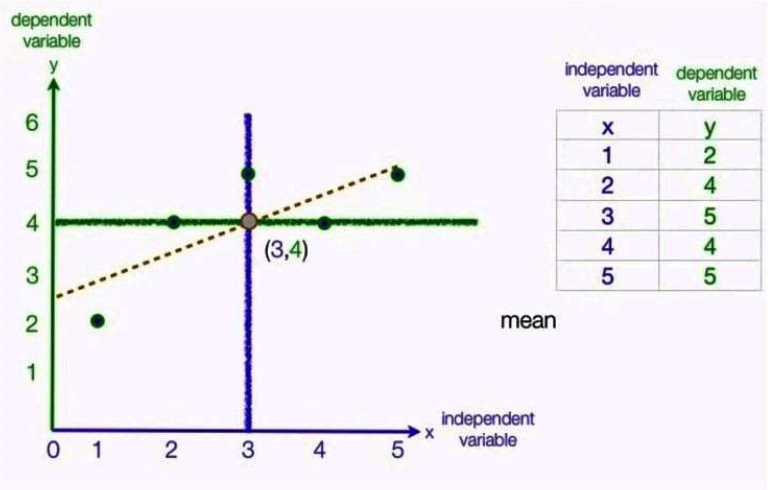
Regression Line Equation is **y = mx + b**

$$m = \frac{\sum(x - \bar{x})(y - \bar{y})}{\sum(x - \bar{x})^2}$$

Linear Regression using Least Square Method



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x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})^2$	$(x - \bar{x})(y - \bar{y})$
1	2				
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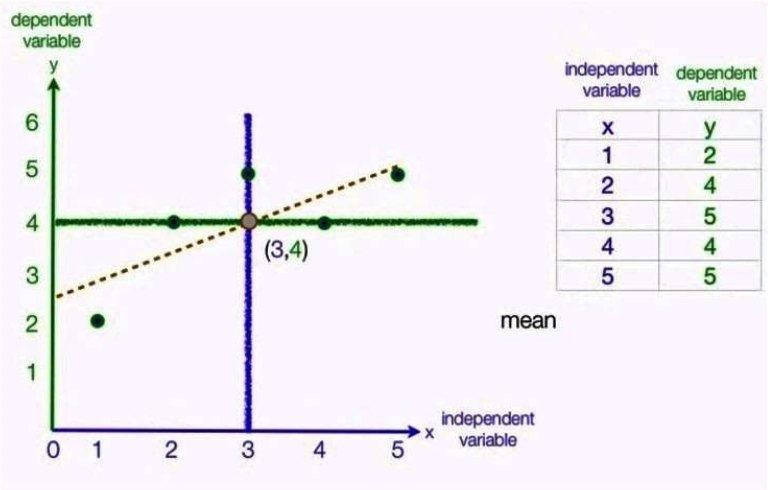

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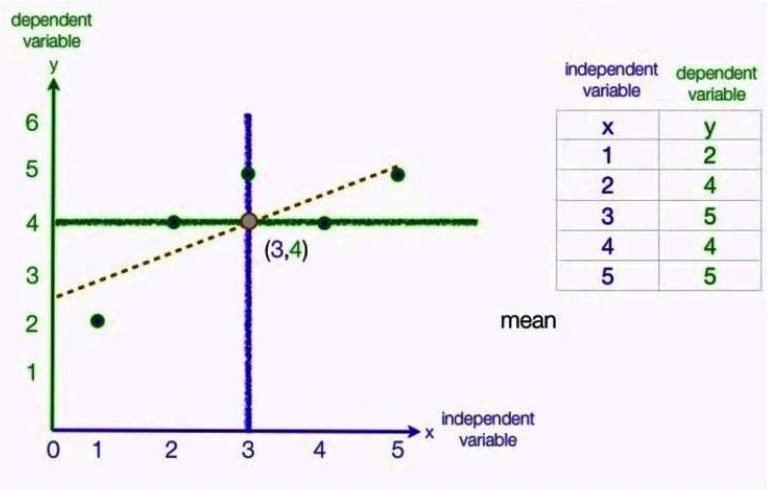
4	4
5	5

x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})^2$	$(x - \bar{x})(y - \bar{y})$
1	2	-2			
2	4	-1			
3	5	0			
4	4	1			
5	5	2			

Linear Regression using Least Square Method



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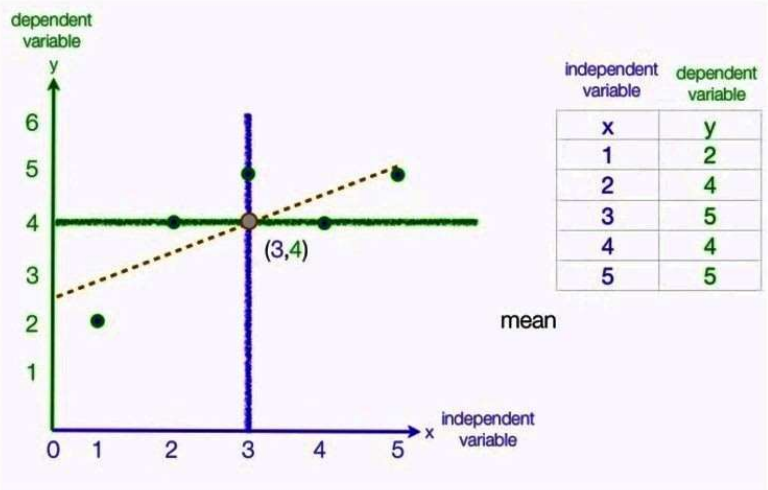
1	2	-2	-2		
2	4	-1	0		
3	5	0	1		
4	4	1	0		
5	5	2	1		

Linear Regression using Least Square Method



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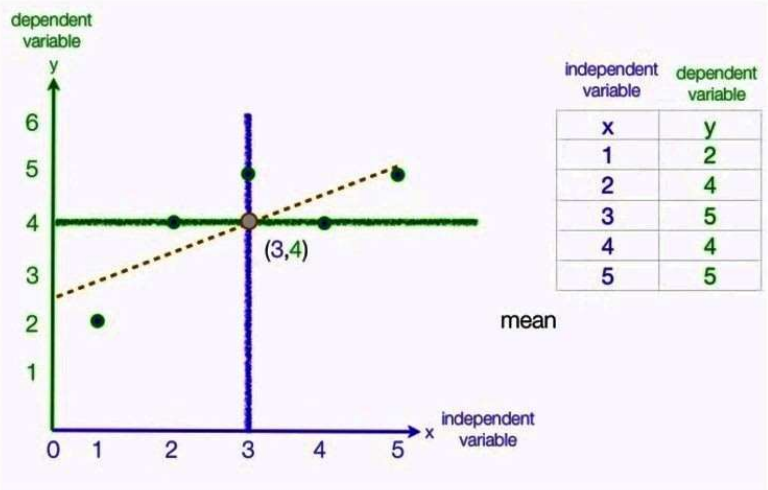
2	4	-1	0	1	
3	5	0	1	0	
4	4	1	0	1	
5	5	2	1	4	

Linear Regression using Least Square Method



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2	4	-1	0	1	0
3	5	0	1	0	0
4	4	1	0	1	0
5	5	2	1	4	2

Linear Regression using Least Square Method

x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})^2$	$(x - \bar{x})(y - \bar{y})$
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2	4	-1	0	1	0
3	5	0	1	0	0
4	4	1	0	1	0
5	5	2	1	4	2

$$\boldsymbol{m} = \frac{\sum(\boldsymbol{x} - \bar{\boldsymbol{x}})(\boldsymbol{y} - \bar{\boldsymbol{y}})}{\sum(\boldsymbol{x} - \bar{\boldsymbol{x}})^2} = \frac{\mathbf{6}}{\mathbf{10}} = \mathbf{0.6}$$

Find the value of **b** from is **y = mx + b using mean values**

$\mathbf{b} = 4 - 0.6*3 = \mathbf{2.2}$

Linear Regression with Squared Loss

In matrix-vector notation, can write it compactly as $\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{w}\|_2^2 = (\boldsymbol{y} - \boldsymbol{X}\boldsymbol{w})^\top(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{w})$

- In this case, the loss function will b

$$L(\boldsymbol{w}) = \sum_{n=1}^{N}(y_n - \boldsymbol{w}^\top \boldsymbol{x}_n)^2$$

- Let us find the $\boldsymbol{w}$ that optimizes (minimizes) the above squared loss

The "least squares" (LS) problem Gauss-Legendre, 18th century)

- We need calculus and optimization to do this!
- The LS problem can be solved easily and has a closed form

$$\boldsymbol{w}_{LS} = \arg\min_{\boldsymbol{w}} L(\boldsymbol{w}) = \arg\min_{\boldsymbol{w}} \sum_{n=1}^{N}(y_n - \boldsymbol{w}^\top \boldsymbol{x}_n)^2$$

$$\boxed{\boldsymbol{w}_{LS} = \left(\sum_{n=1}^{N} \boldsymbol{x}_n \boldsymbol{x}_n^\top\right)^{-1}\left(\sum_{n=1}^{N} y_n \boldsymbol{x}_n\right) = (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top \boldsymbol{y}}$$

Link to a nice derivation

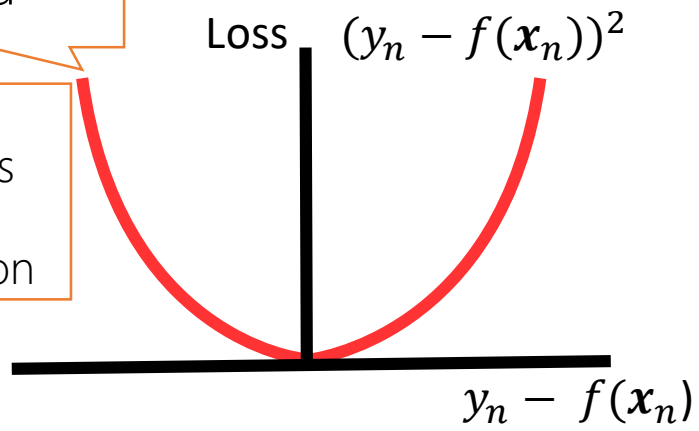
Alternative loss functions

- Many possible loss functions for regression prob

Choice of loss function usually depends on the nature of the data. Also, some loss functions result in easier optimization problem than others

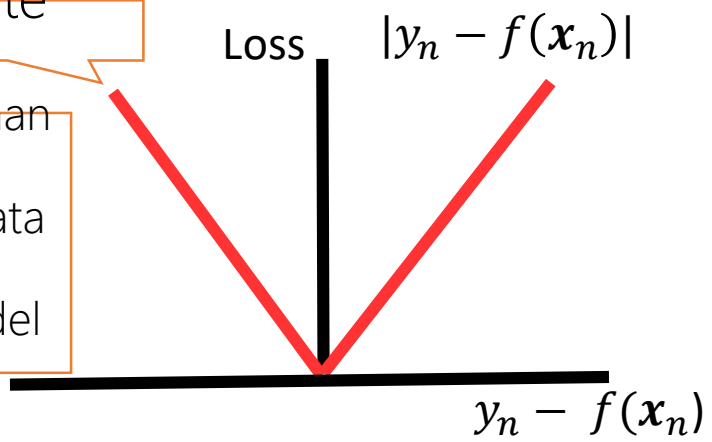
Squared loss

Very commonly used for regression. Leads to an easy-to-solve optimization problem



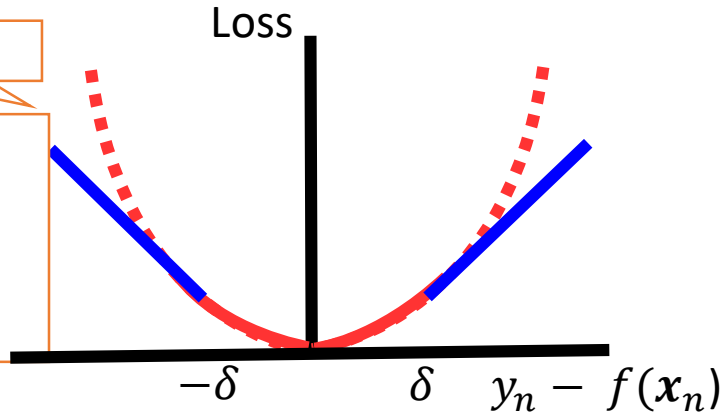
Absolute loss

Grows more slowly than squared loss. Thus better suited when data has some outliers (inputs on which model makes large errors)



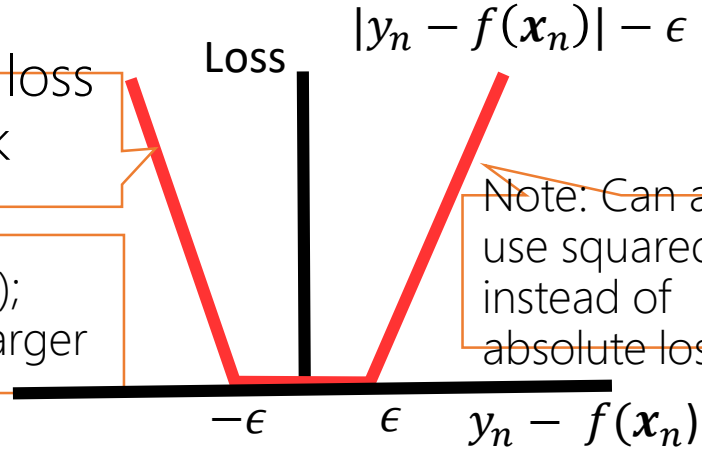
Huber loss

Squared loss for small errors (say up to δ); absolute loss for larger errors. Good for data with outliers



ϵ-insensitive loss (a.k.a. Vapnik loss)

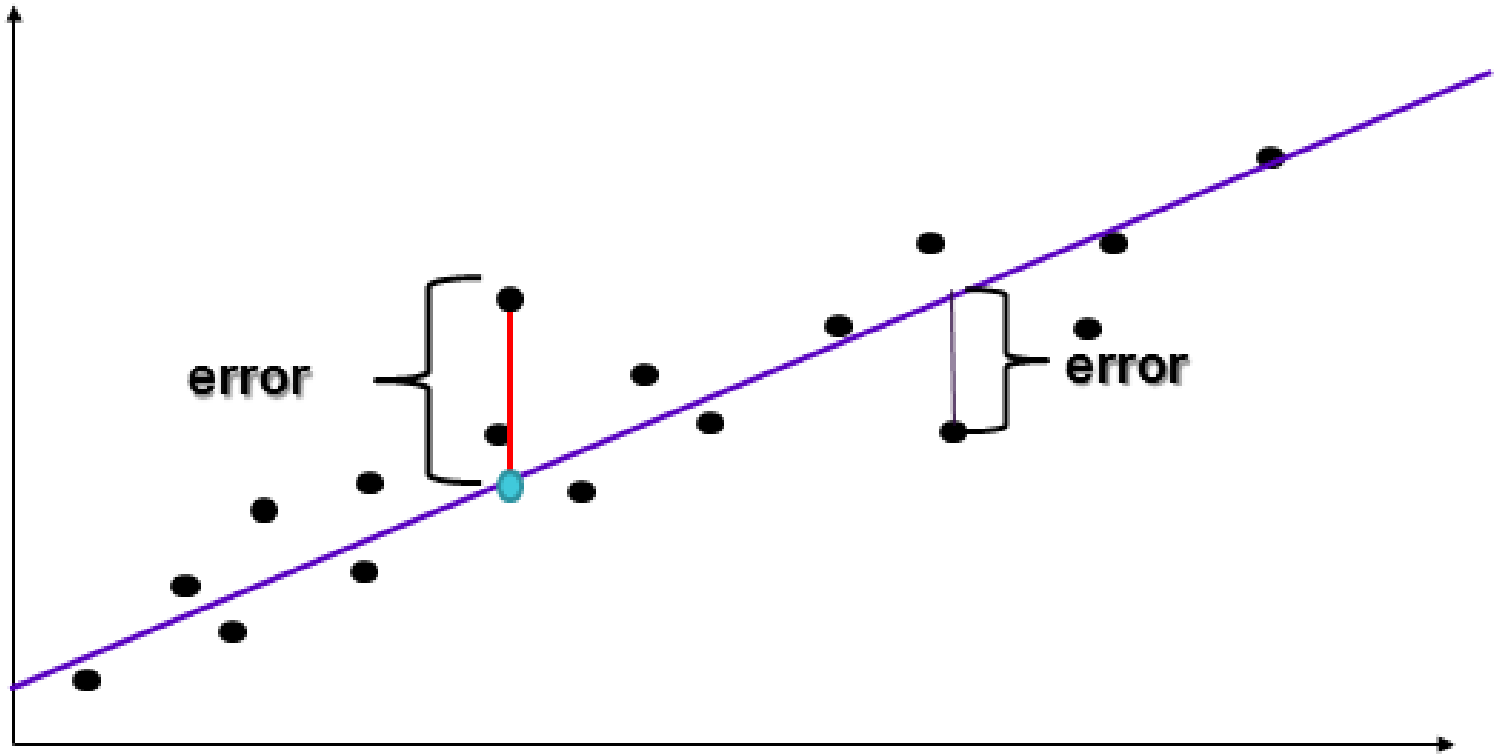
Zero loss for small errors (say up to ϵ); absolute loss for larger errors



Note: Can also use squared loss instead of absolute loss

Linear Regression Performance Evaluation

Error Variable



The inclusion of the random error term allows (x,y) to fall either above the true regression line (When E > 0) or below the line (When E < 0)

Performance Evaluation

- The performance of a regression model can be understood by knowing the error rate of the predictions made by the model.
- One can also measure the performance by knowing how well your regression line fit the dataset.
- A good regression model is one where the difference between the actual or observed values and predicted values for the selected model is small and unbiased for train, validation and test data sets.

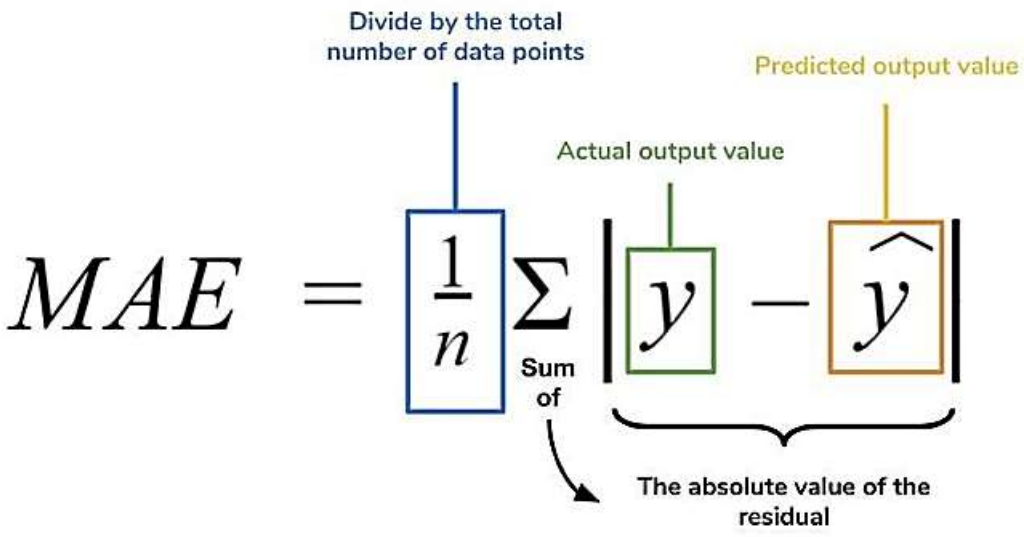
Performance Evaluation

To measure the performance of your regression model, some statistical metrics are used. Here we will discuss four of the most popular metrics. They are-

- Mean Absolute Error(MAE)
- Root Mean Square Error(RMSE)
- Coefficient of determination or R2
- Adjusted R2

Mean Absolute Error(MAE)

This is the simplest of all the metrics. It is measured by taking the average of the absolute difference between actual values and the predictions.



Root Mean Square Error (RMSE)

- The Root Mean Square Error is measured by taking the square root of the average of the squared difference between the prediction and the actual value.
- It represents the sample standard deviation of the differences between predicted values and observed values(also called residuals).

$$RMSE = \sqrt{\frac{\sum_{i=1}^{N} (Predicted_i - Actual_i)^2}{N}}$$

MAE vs. RMSE

- Compared to MAE, RMSE gives a higher total error and the gap increases as the errors become larger. It penalizes a few large errors more than a lot of small errors. If you want your model to avoid large errors, use RMSE over MAE.
- Root Mean Square Error (RMSE) indicates the average error in units of y, the predicted feature, but penalizes larger errors more severely than MAE. A value of 0 indicates a perfect fit.
- You should also be aware that as the sample size increases, the accumulation of slightly higher RMSEs than MAEs means that the gap between these two measures also increases as the sample size increases.

R^2 Error

- It measures how well the actual outcomes are replicated by the regression line.
- It helps you to understand how well the independent variable adjusted with the variance in your model.
- That means how good is your model for a dataset. The mathematical representation for R^2 is –

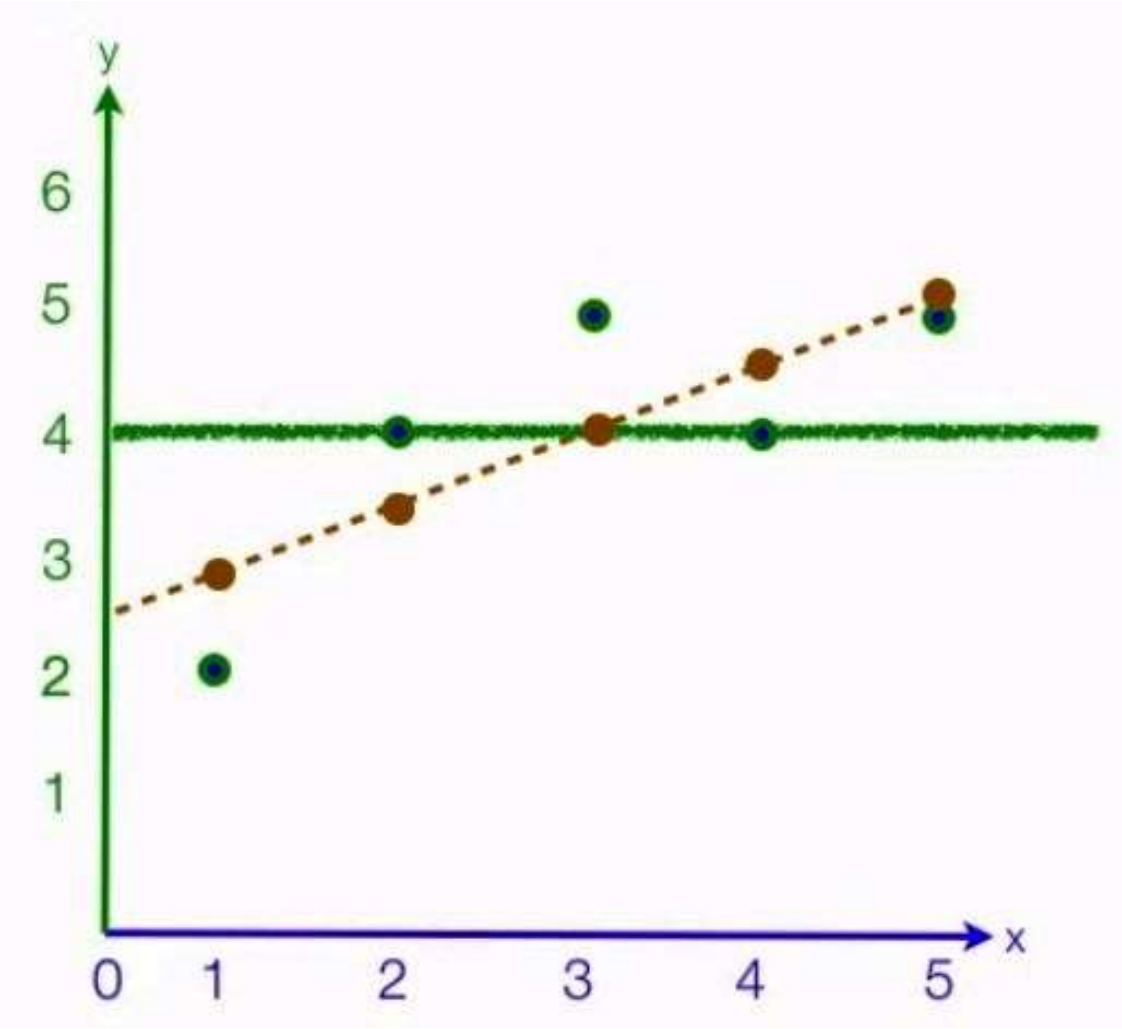
$$R^2 = \frac{SSR}{SST} = \frac{\sum(\hat{y}_i - \bar{y})^2}{\sum(y_i - \bar{y})^2}$$

Here,

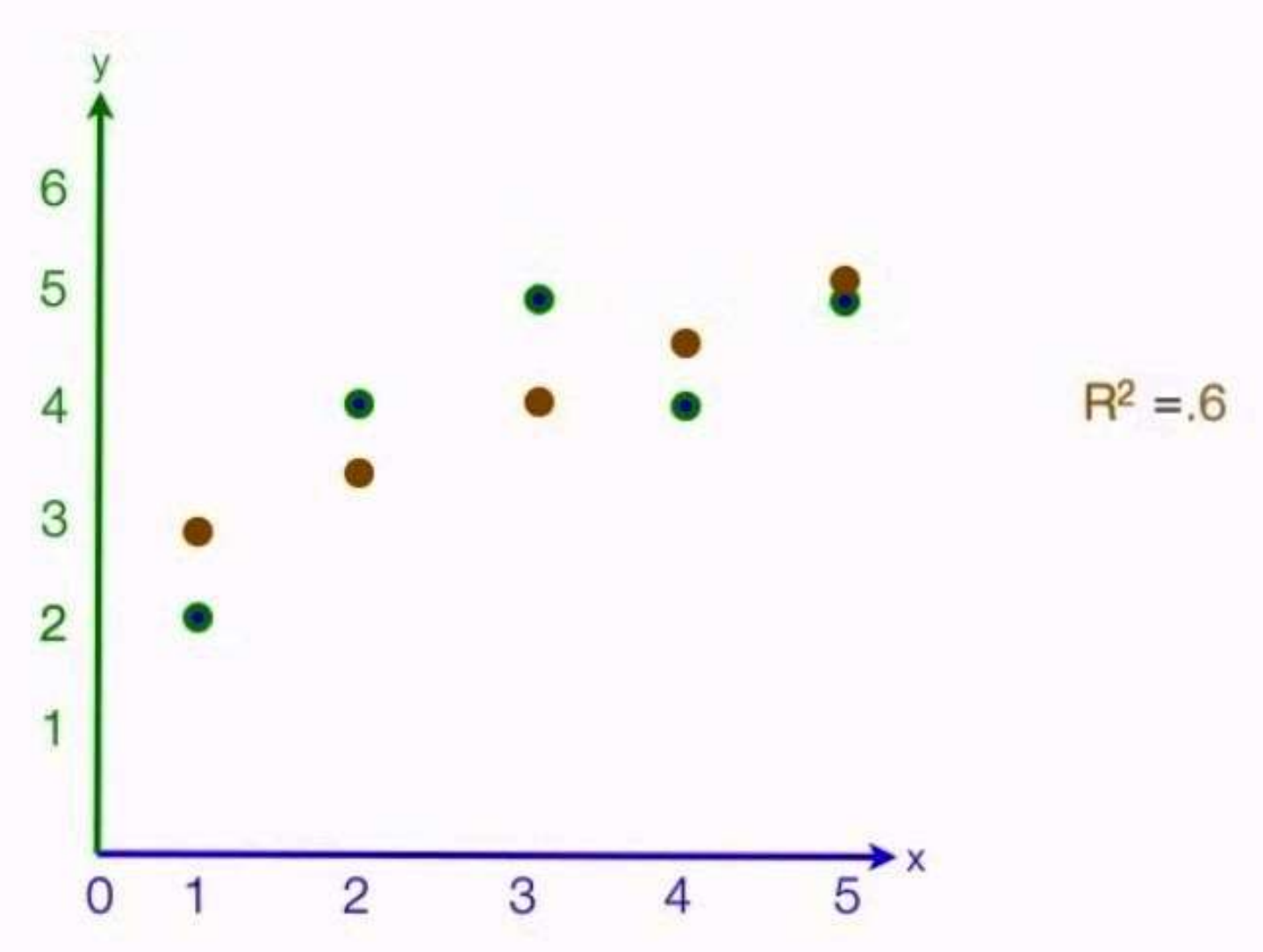
– SSR = Sum Square of Residuals(the squared difference between the predicted and the average value)

– SST = Sum Square of Total(the squared difference between the actual and average value)

R-Squared ($\mathbf{R^2}$)



R-Squared ($\mathbf{R^2}$)



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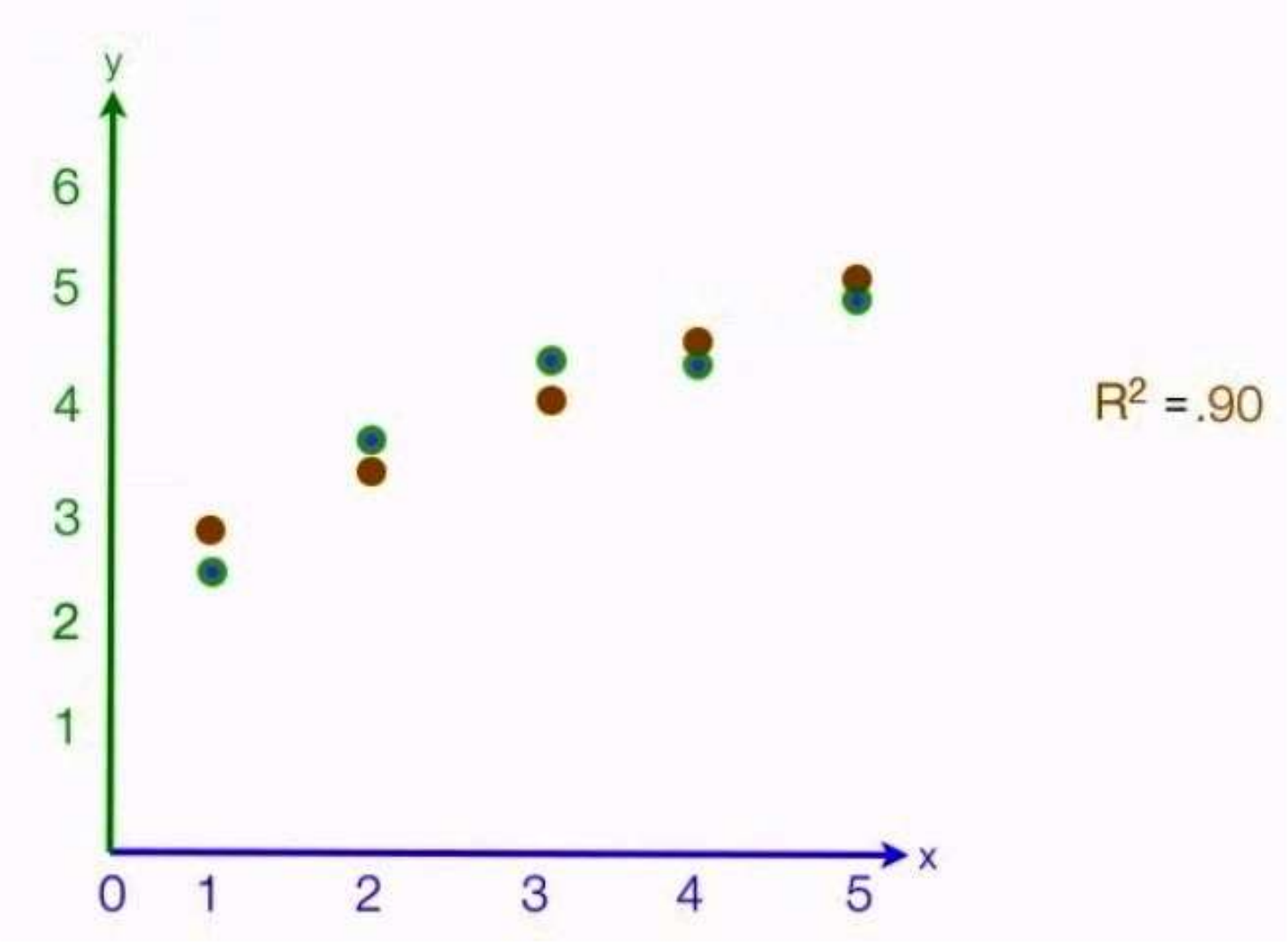

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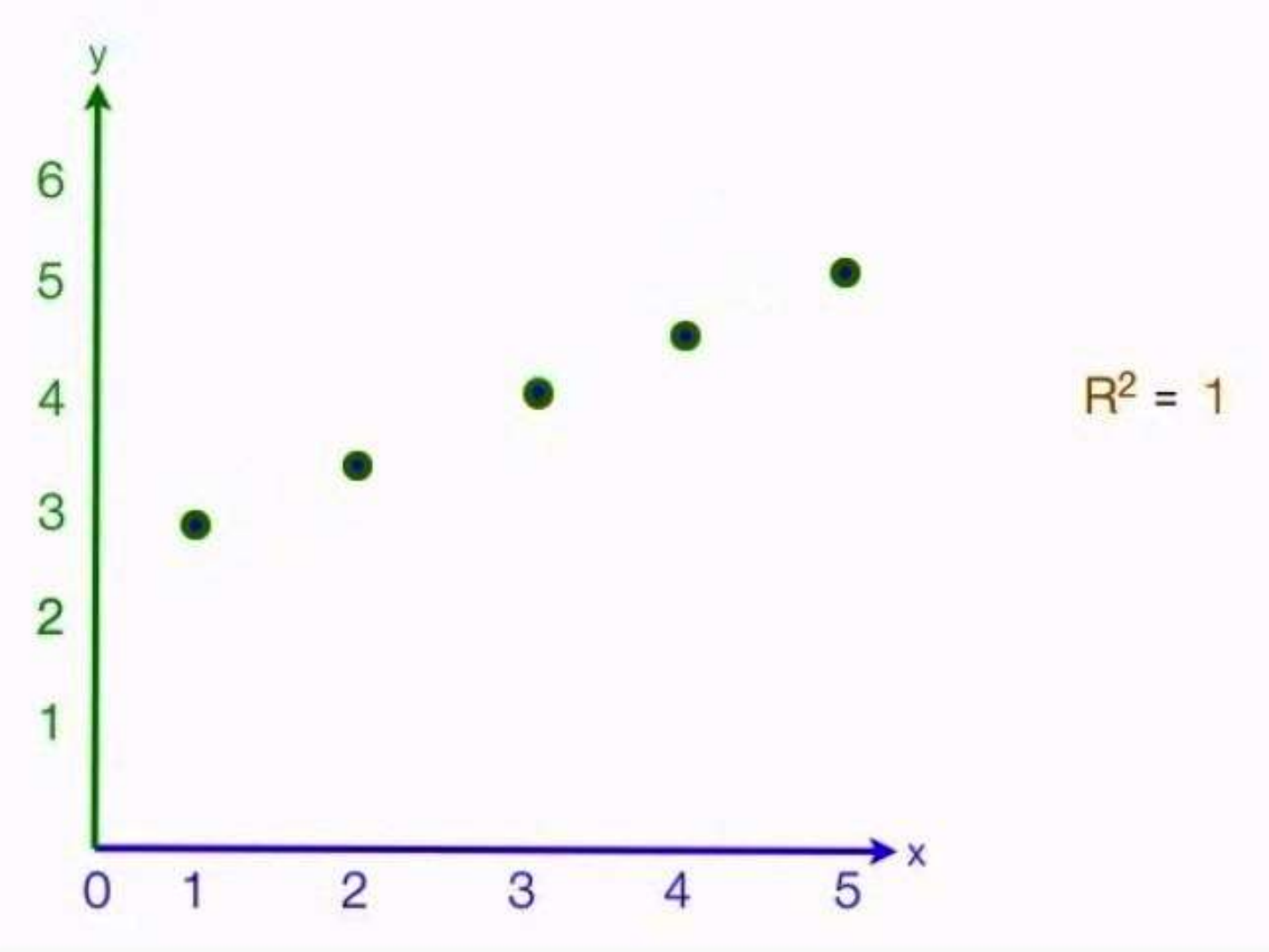

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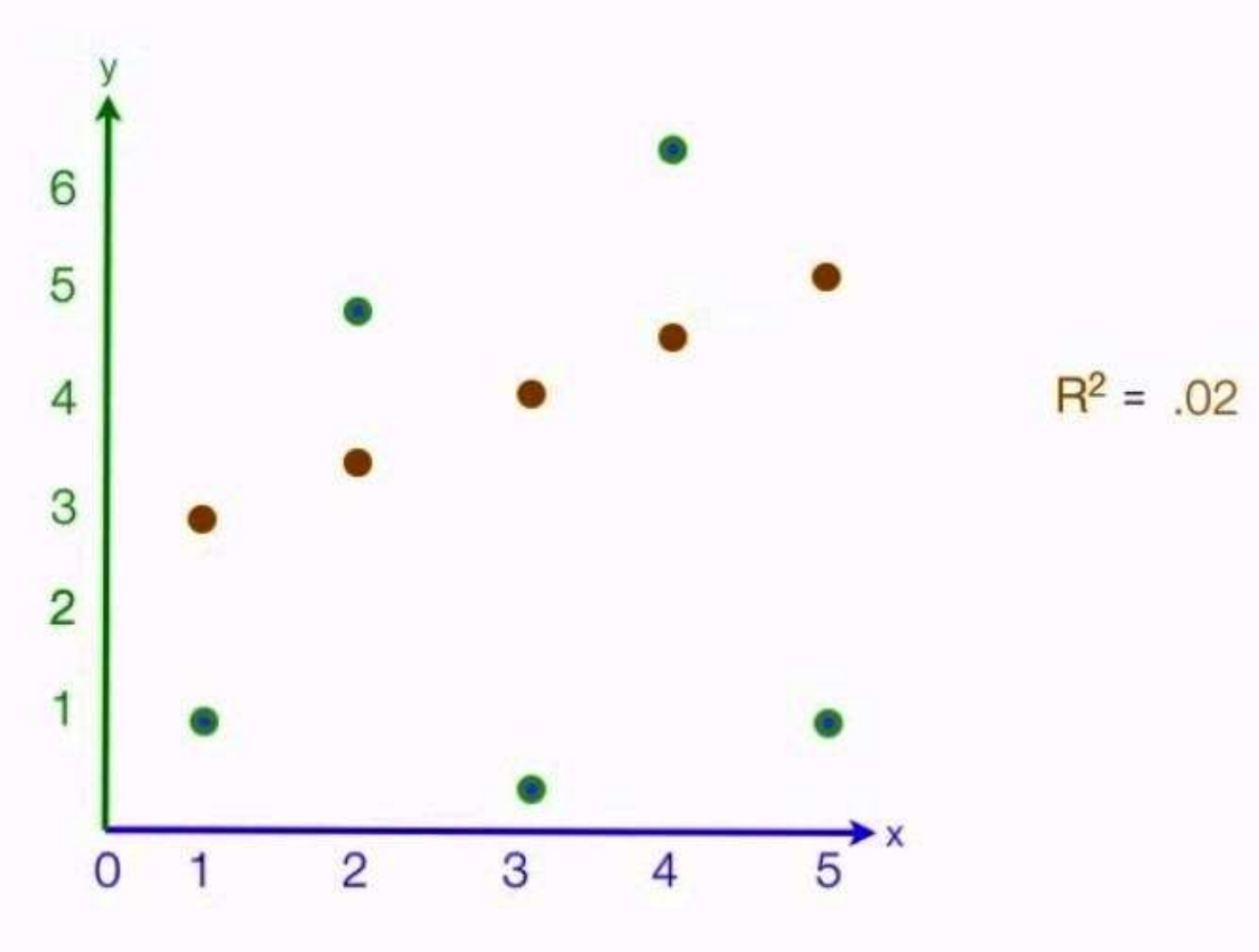

Adjusted R^2 Error

- Modified form of R-Squared whose value increases if new predictors tend to improve models performance and decreases if new predictors does not improve performance as expected.
- The Adjusted R-squared would penalize you for adding features which are not useful for predicting the target.
- It takes into account the number of independent variables used for predicting the target variable.

$$Adjusted\ R^2 = 1 - \frac{(1 - R^2)(N - 1)}{N - p - 1}$$

where,

N = number of records in the data set.

p = number of independent variables.

LR Exercise 1

Q. Study the relationship between the monthly sales and the advertising costs surveyed for different stores as given below. Find the equation of the straight line that fits the data best. Determine the R-squared value.

Store	Sales (units)	Advertising Cost
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LR Exercise 2

Q. Examine the relationship between the age and price for used cars sold in the last year by a car dealership company. Find the equation of the straight line that fits the data best. Determine the R-squared value.

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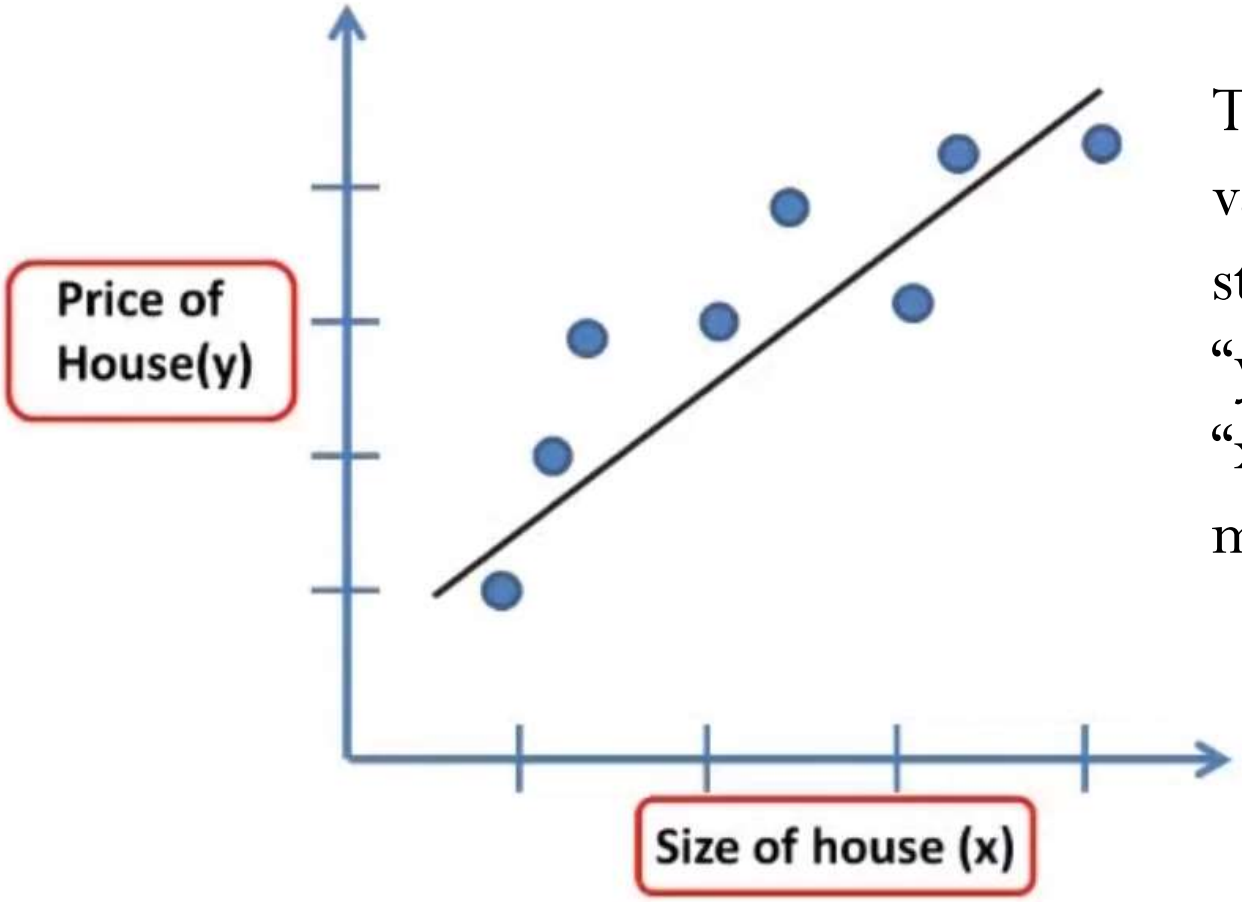
Gradient Descent in LR



The linear relationship between 2 variables can be represented as a straight line **y = mx + b**

"y" is target

"x" is feature

m & b are model parameters

Gradient Descent in LR

A general linear equation with multiple features can be written as:

$$\boldsymbol{y = b + m_1 x_1 + m_2 x_2 + \ldots\ldots\ldots + m_n x_n}$$

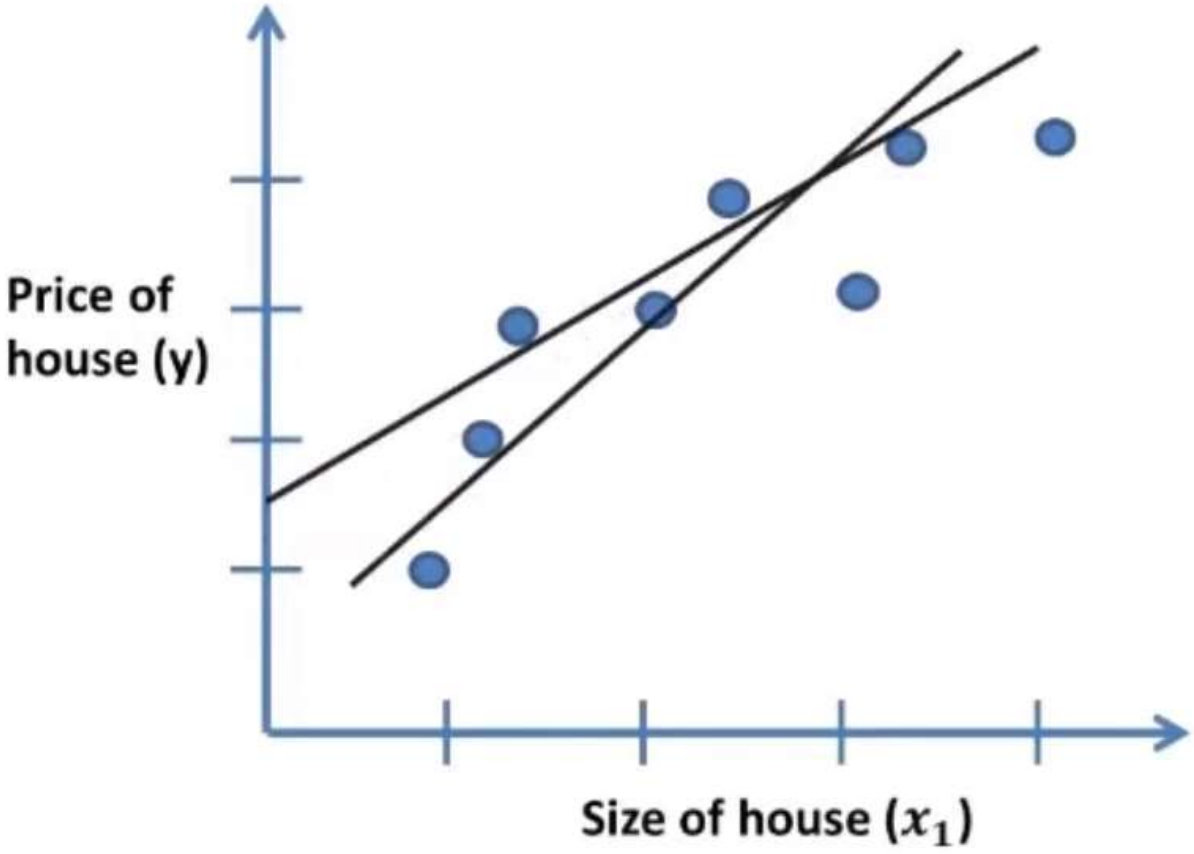
xi are features

b, mi are model parameters or coefficient

y is target variable

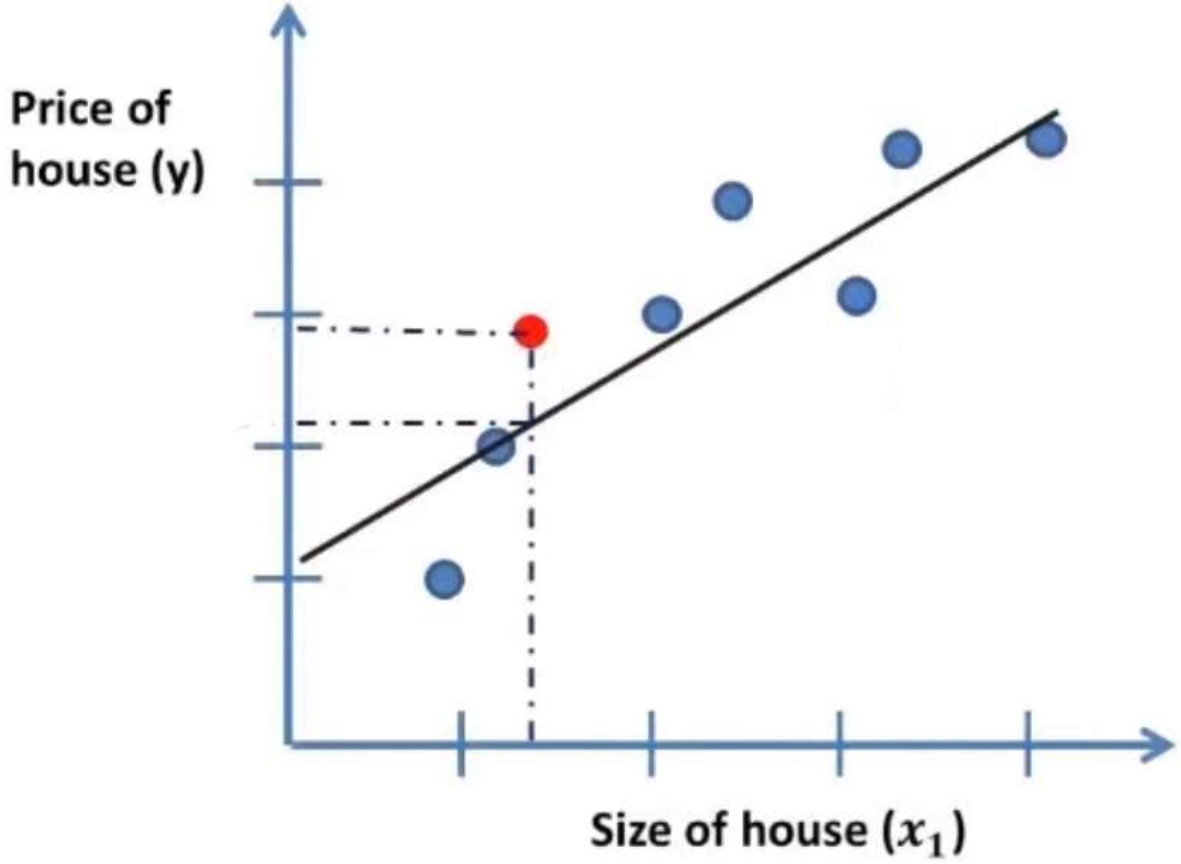
Gradient Descent in LR

Depending on the values of m and b, multiple possible lines can be possible.



We need to find out the value of parameters **b** and **m** corresponding to which straight line fit best to the data.

Gradient Descent in LR



- **Error**
  $e = \text{actual} - \text{predicted}$
  $e = y - \hat{y}$

- For i samples we will have e_i error values.

- e_i can be +ve or –ve, therefore take square of it.

The Cost Function of Linear Regression

- Cost function measures how a machine learning model performs.
- Cost function is the calculation of the error between predicted values and actual values, represented as a single real number.
- Cost function of a linear regression is mean square error.
  - Cost Function $\mathbf{J} = \frac{\sum(\widehat{y_i} - y_i)^2}{n}$

    After expanding $\hat{y} = \mathbf{b} + \mathbf{mx}$ in above equation we get

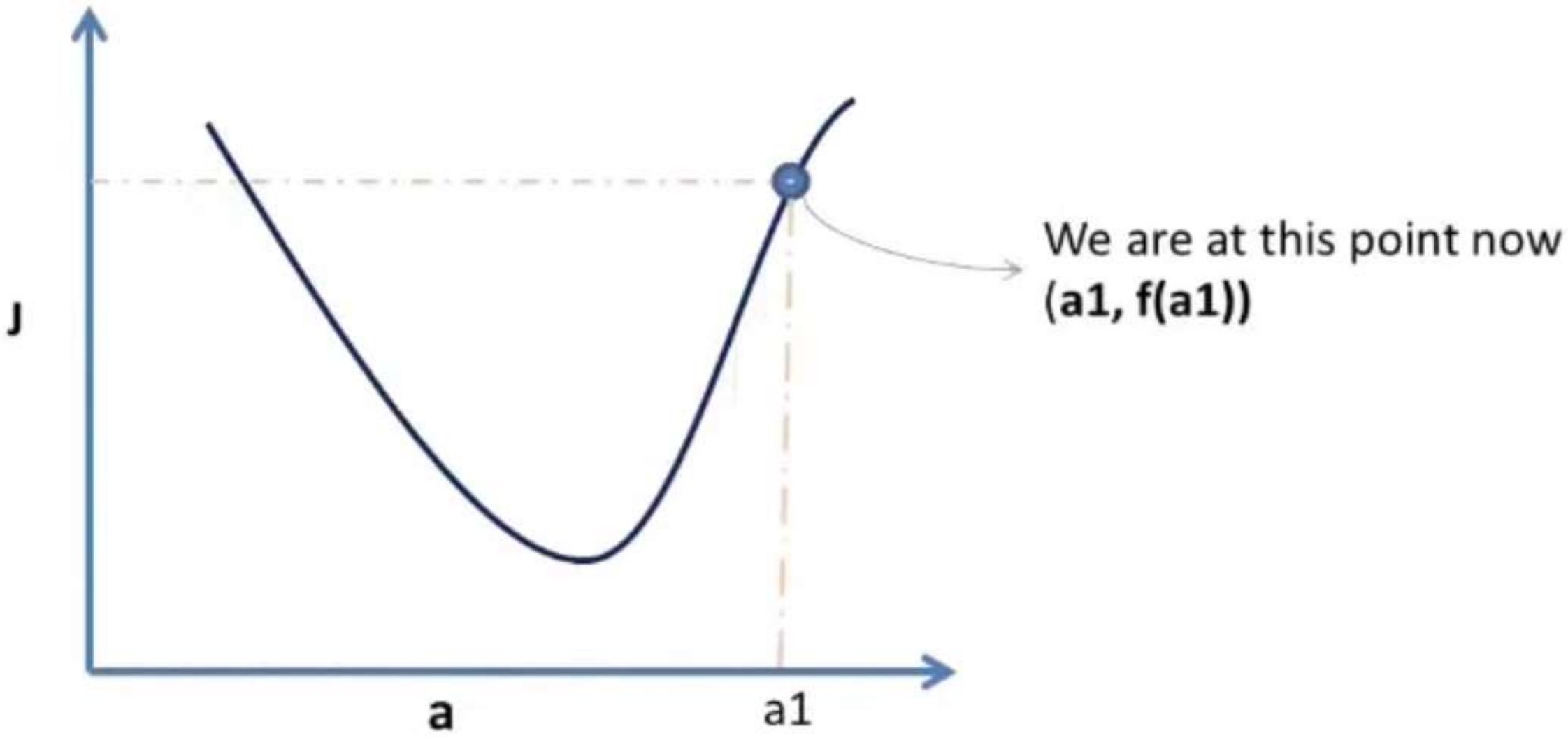
$$J = \frac{\sum(b + m_i x_i - y_i)^2}{n}$$

- Therefore the best fitting model would be the one which minimizes the value of cost function.

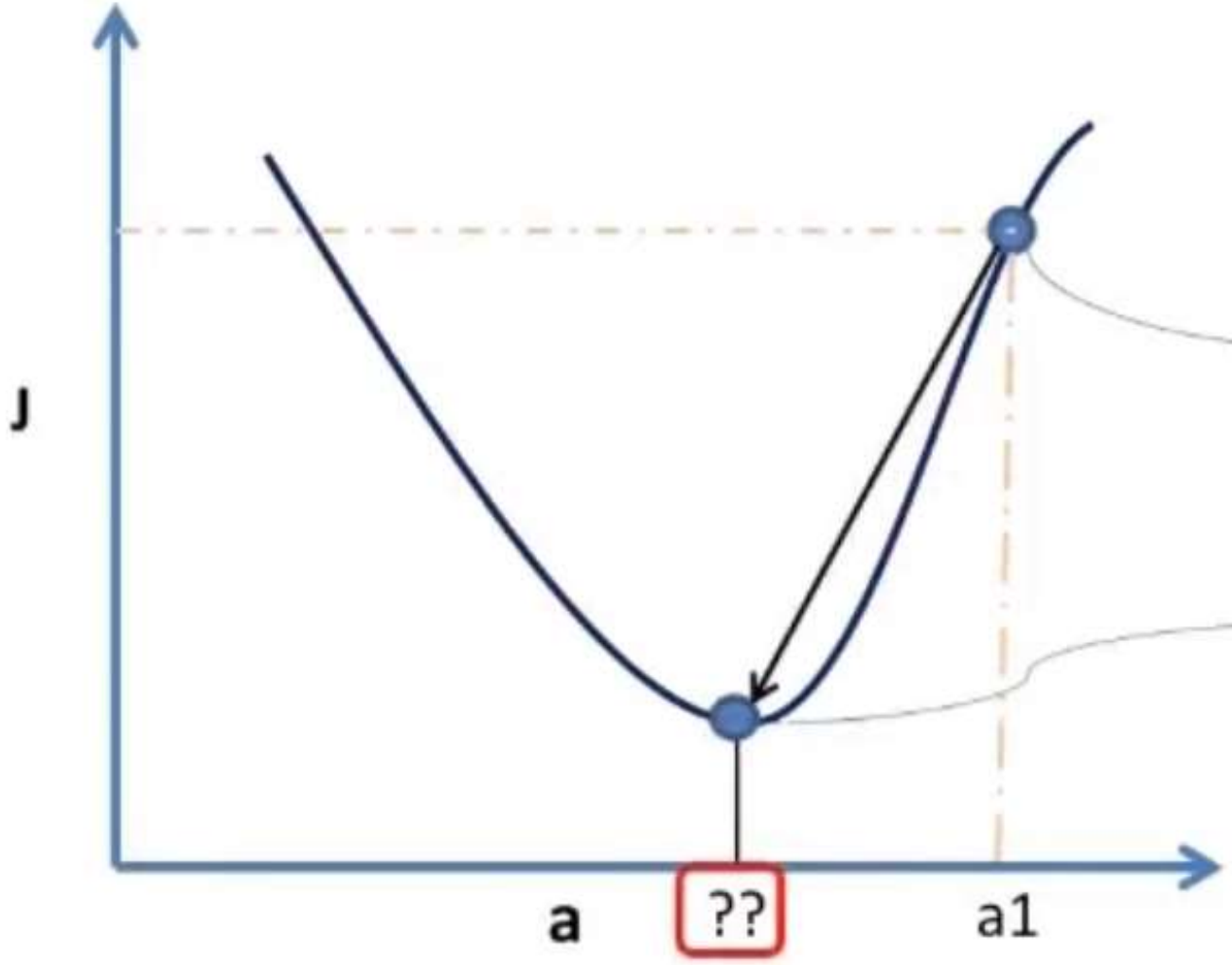
How to minimize the Cost Function?

- We have established the fact that all the straight lines are just different combination of model parameters b & m.
- Cost Function is the function of parameters **b** & **m**.
- Therefore by changing the values of b & m we can change the **cost function.**
- We will keep changing the values of b & m till we find a combination where cost function is **minimized**.
- To find the best combination we use **Gradient Descent Algorithm**.

Gradient Descent in LR



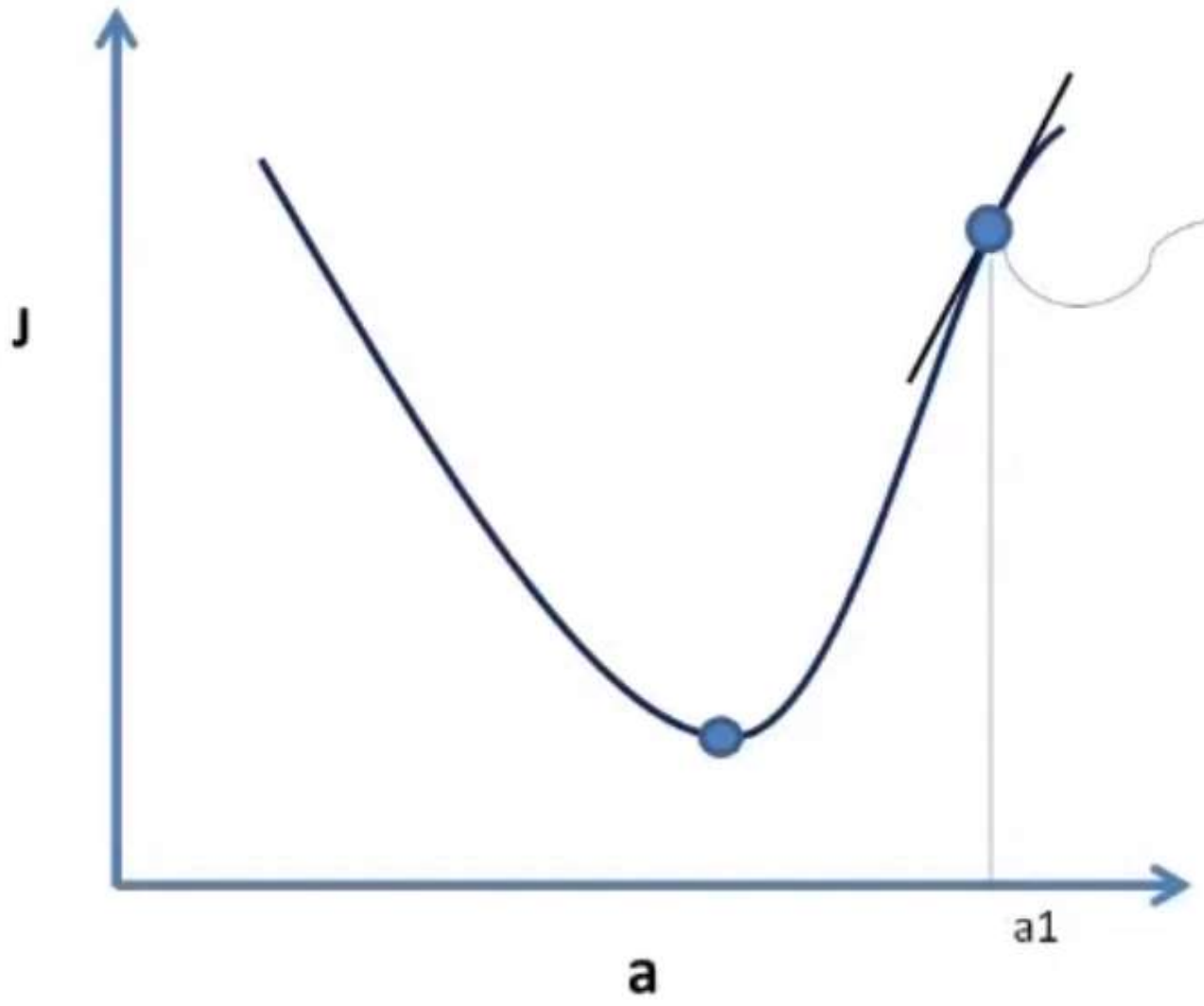
Gradient Descent in LR



We are at this point now **(a1, f(a1))**

We want to reach here, we want to know **for what value of 'a'** our **J = f(a) will assume minimum value**.

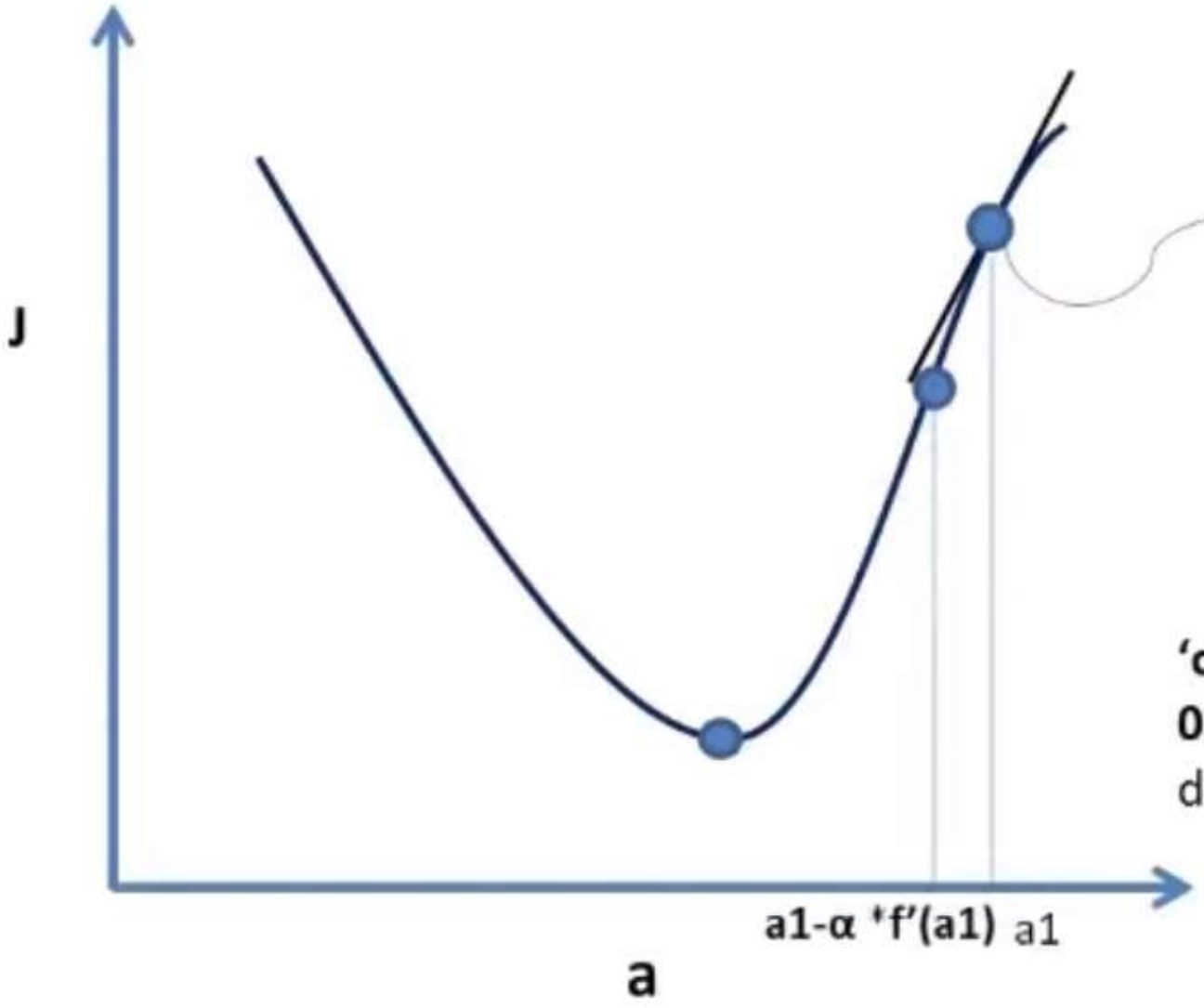
Gradient Descent in LR



Lets calculate slope at this point.

It is $\frac{dJ}{da}$ **at a1** or f'(a1)

Gradient Descent in LR



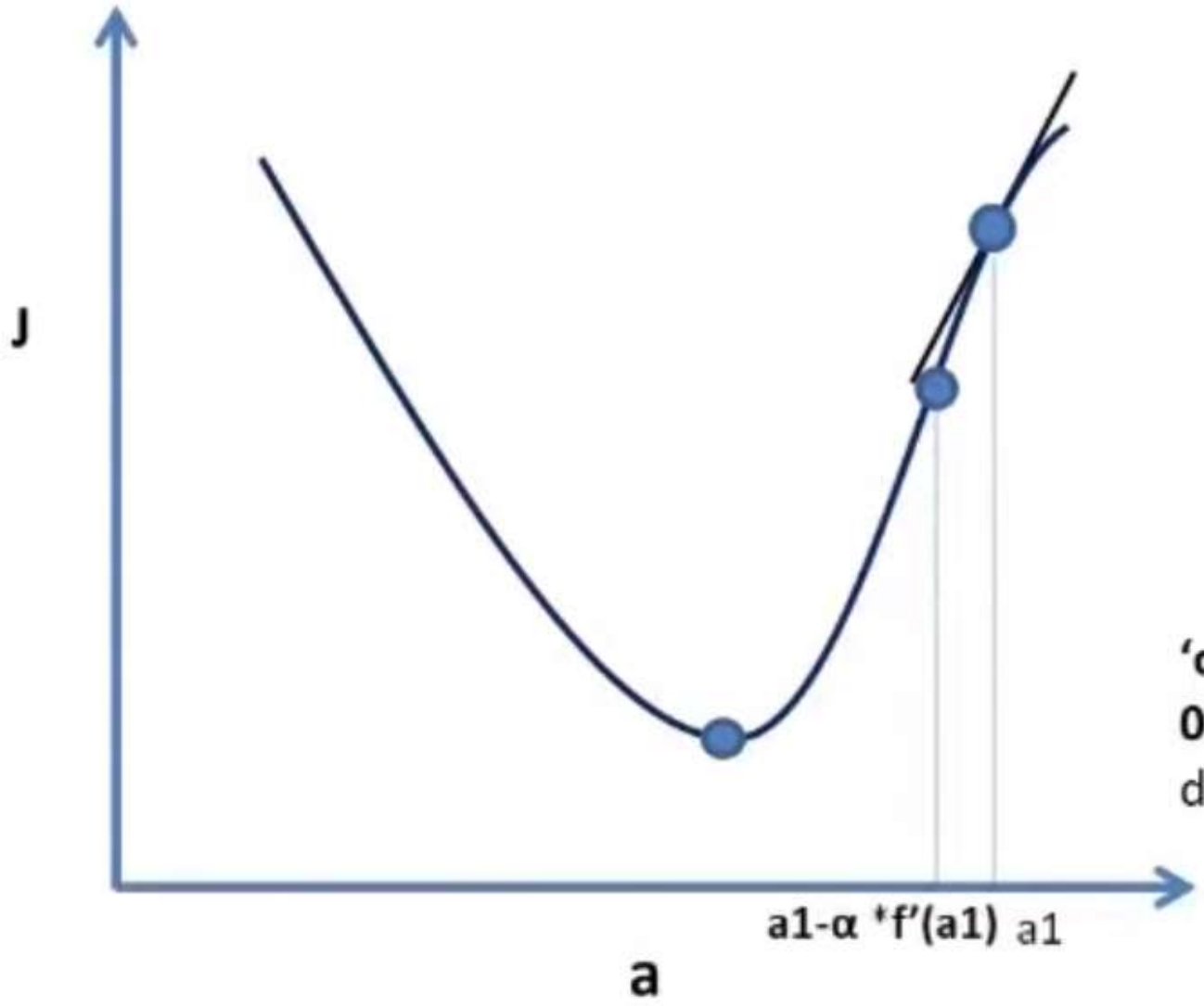
Lets calculate slope at this point.

It is $\frac{dJ}{da}$ **at a1** or f'(a1)

Lets move a step **'α'** in this direction to reach **a1- α *f'(a1)**

'α' is a small fixed quantity in the range of **0.01**. Therefore, the size of our steps is dependent on the slope **f'(a1)**.

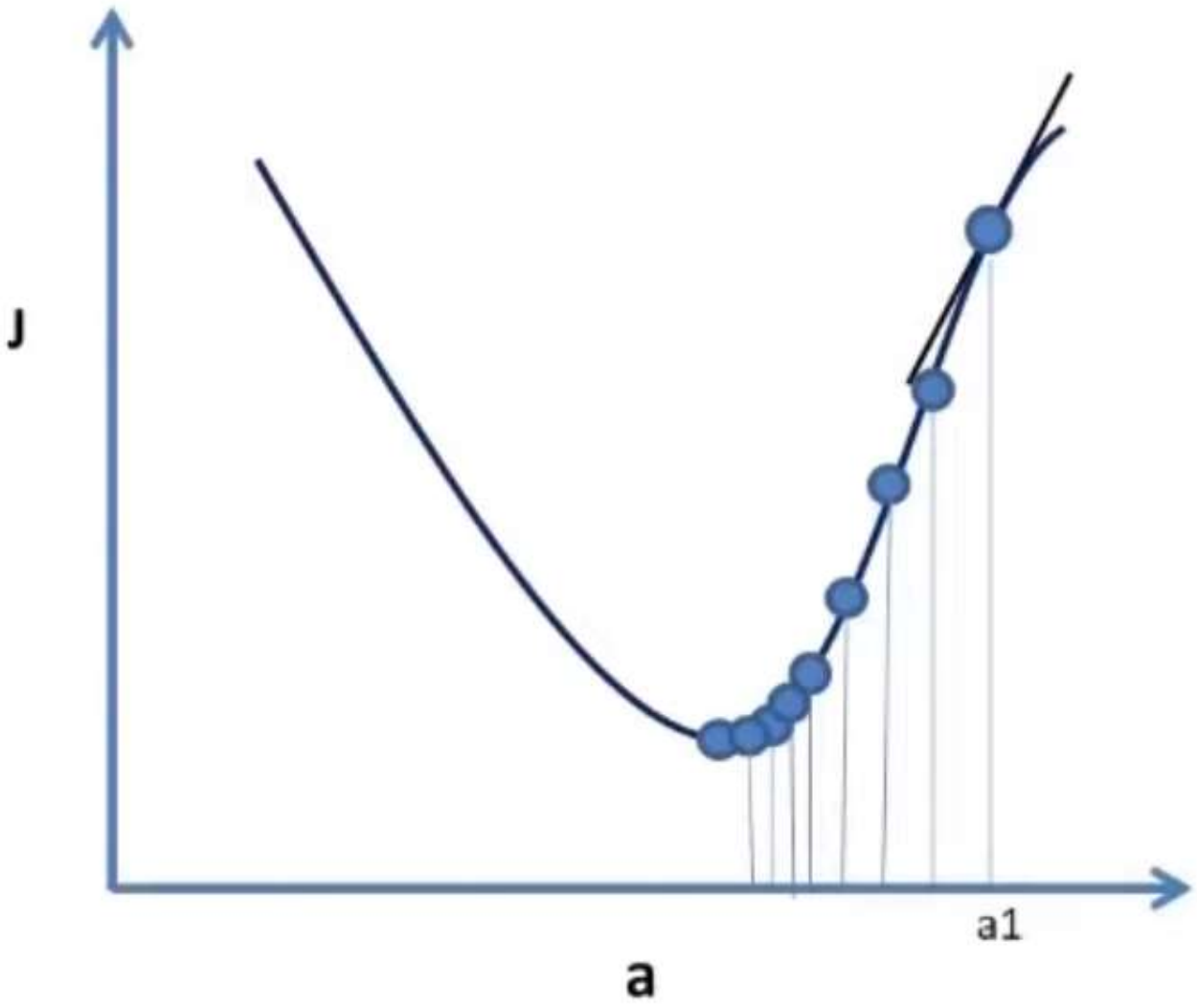
Gradient Descent in LR



Higher the value of slope (which occurs away from minima) **larger the steps and vice versa.**

'α' is a small fixed quantity in the range of **0.01**. Therefore, the size of our steps is dependent on the slope **f'(a1)**.

Gradient Descent in LR



As we move closer to minimum, **the slope decreases** and hence our **steps towards minimum keeps on getting smaller**. At minimum, slope becomes **zero**...

Gradient Descent in LR

- These iteration of Gradient Descent algorithm can run multiple times depending on nature of function and 'α' (learning rate) and of course where we start from, 'a1' in this case.

- Same methodology can be used to minimize cost function (J) which is a function of model parameters b & m by changing them through iterations of Gradient Descent Algorithm.

Gradient Descent in LR

Steps

1. Calculate slope at the current value of parameter b & m separately
2. Take step α and update new parameters.
3. Calculate the cost function J with new (b & m) values.

Repeat it multiple times.

What happen if take the larger value of α??

Types of Regression Analysis

Multiple Regression

Definition and Concept
Multiple regression involves the prediction of a dependent variable through several independent variables, offering a more nuanced understanding of how various factors collectively influence outcomes, thereby capturing complex interdependencies.

Mathematical Representation
The mathematical form of multiple regression is expressed as $Y = a + b1X1 + b2X2 + \ldots + bnXn$, where Y is the dependent variable, a is the intercept, b represent the coefficients of the independent variables, and X are the respective predictors.

Differences from Linear Regression
Unlike simple linear regression, which focuses on a single predictor, multiple regression considers multiple predictors simultaneously, allowing for the exploration of interaction effects and multivariate relationships that simpler models may overlook.

Multiple Regression – Key Components

Multiple Predictor Variables
Utilization of multiple predictor variables enables a detailed analysis of how different factors contribute to the outcome variable, yielding richer insights than single-variable analysis alone.

Interaction Effects
Multiple regression can identify interaction effects, where the influence of one predictor variable on the dependent variable differs depending on the level of another predictor, enhancing the model's explanatory power.

Model Fit and Evaluation
Assessing model fit entails evaluating how well the regression model approximates real data, commonly measured using R-squared, adjusted R-squared, and residual analysis to determine the robustness of the model's predictions.

Logistic Regression

Definition
Logistic regression is a statistical technique used to model binary outcome variables based on one or more predictor variables, applying a logistic function to predict the probability of a specified outcome.

Binary Outcomes
Unlike linear regression, which predicts continuous outcomes, logistic regression focuses on binary outcomes, such as yes/no, true/false, or success/failure, making it essential for scenarios requiring dichotomous decisions.

Applications in Healthcare and Marketing
Logistic regression is extensively utilized in healthcare for predicting disease outcomes based on risk factors and in marketing for determining the likelihood of a customer purchasing a product based on various influencing factors.

Polynomial Regression

Definition
Polynomial regression is a form of regression analysis in which the relationship between the independent variable and the dependent variable is modeled as an nth degree polynomial, providing flexibility to accommodate curves and complex relationships.

Non-linear Relationships
This regression type is advantageous for scenarios where relationships are not adequately captured by linear models, allowing researchers to fit polynomial functions that align more closely with the observed data patterns.

Use Cases in Engineering and Environmental Science
Polynomial regression finds significant applications in engineering for tasks such as optimization problems and simulation modeling, as well as in environmental science for analyzing complex relationships among ecological variables.

Ridge and Lasso Regression

Definition
Ridge and Lasso regression are techniques used to prevent overfitting by applying regularization to linear regression models. Ridge regression adds a penalty based on the size of coefficients, while Lasso regression can reduce coefficients to zero, effectively selecting variables.

Regularization Techniques
These techniques help maintain a balance between model complexity and predictive accuracy, enabling models to perform better on unseen data and reducing the risks associated with multicollinearity among predictors.

Applications in Machine Learning and Data Science
Ridge and Lasso regression are essential tools in machine learning for feature selection and improving predictive models. They are frequently used in various domains, including finance, bioinformatics, and marketing analyses.

Multivariate Regression

Definition
Multivariate regression refers to the statistical process that models the relationship between multiple independent and several dependent variables, seeking to understand the impact of various predictors on outcomes.

Multiple Predictors
This method allows for the integration of complex data inputs, assessing how several factors simultaneously influence the dependent variables, thus yielding richer insights than univariate models.

Applications in Social Sciences and Marketing
Multivariate regression is crucial in social sciences for analyzing influential factors of behaviors and trends, while in marketing, it helps understand purchasing behaviors across multiple demographics and influences.

Multiple Linear Regression

Multiple Linear Regression

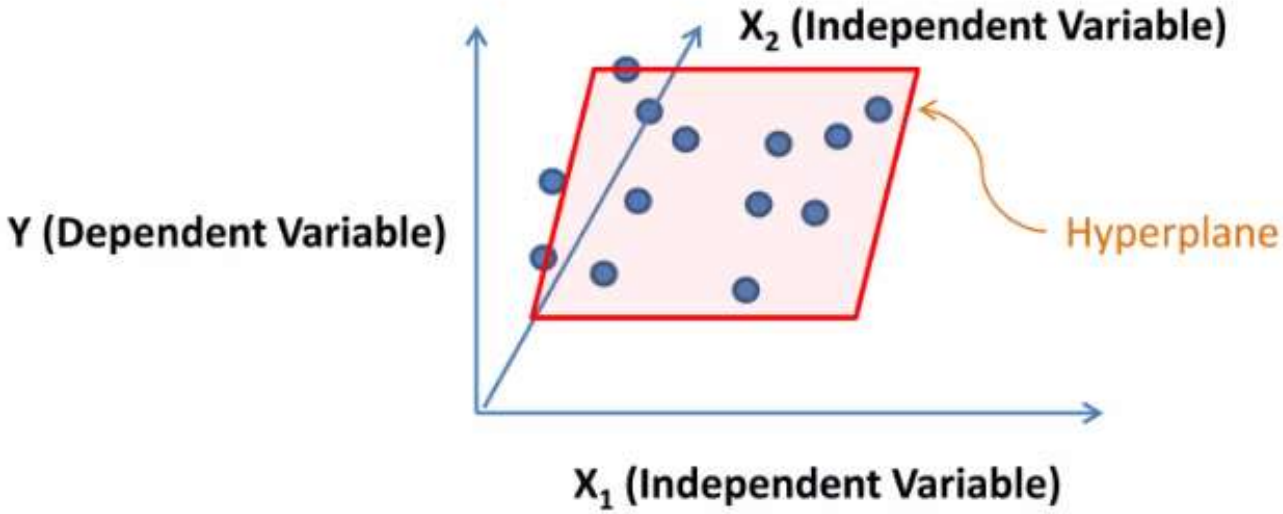
- Multiple linear regression is an extension of simple linear regression, which models the relationship between a dependent variable and multiple independent variables.
- In multiple linear regression, there are multiple independent variables.

Multple Linear Regression

- **Used to Predict a correlation between more than one independent variables and a dependent variable.**

  e.g., Income and Age is correlated with Spending

- **When the data is plotted on a graph, there appears to be a hyperplane relationship.**



Multiple Linear Regression

- The general form of a multiple linear regression model with k independent variables is given by:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_k X_k + \varepsilon$$

Here,

- Y is the dependent variable.
- β_0 is the intercept (the value of Y when all independent variables are zero).
- $\beta_1, \beta_2, \ldots, \beta_k$ are the coefficients associated with each independent variable $X_1, X_2, \ldots, X_k$, representing the strength and direction of their impact on Y.
- $X_1, X_2, \ldots, X_k$ are the independent variables.
- ε represents the error term, which accounts for unobserved factors affecting Y.

How to use Multipls Regression Analysis?

The processes involved the selection of features, engineering the features, feature normalization, selection loss functions, hypothesis analysis, and creating a regression model.

1. **Selection of features:**

It is the most important step in multiple regression. Also known as variable selection, this process involves selecting viable variables to build efficient models.

Feature Elimination

- **In a dataset, we may not want to keep all the independent variables (features) in our model:**

  - **More Features = More Complex Model**
  - **If feature does not contribute to prediction, adds noise to the model.**

ID fields are like random numbers – do not contribute to prediction.

ID	Income	Age	Height	Spending
37	18000	18	5'8"	
38	75000	40	5'9"	
39	27000	28	6'1"	
40	24000	26	5'6"	
41	45000	34	6'2"	

Height not likely or very little to influence spending

How to use Multiple Regression Analysis?

2. Feature Normalization: This involves feature scaling to maintain streamlined distribution and data ratios. This helps in better data analysis. The value of all the features can be changed according to the requirement.

3. Selecting Loss function and hypothesis: The loss function is used for predicting errors. The loss function comes into play when the hypothesis prediction changes from the actual figures. Here, the hypothesis represents the value predicted from the feature or variable.

4. Fixing hypothesis parameter: The parameter of the hypothesis is fixed or set in such a way that it minimizes the loss function and enhances better prediction.

How to use Multiple Regression Analysis?

5. **Reducing the loss function**: The loss function is minimized by generating an algorithm specifically for loss minimization on the dataset which in turn facilitates the alteration of hypothesis parameters.

Gradient descent is the most commonly used algorithm for loss minimization.

6. Analyzing the hypothesis function: The function of the hypothesis needs to be analyzed as it is crucial for predicting the values. After the function is analyzed, it is then tested on test data.

Assumptions in the Multiple Regression Model

- The dependent and the independent variables have a linear relationship.
- The independent variables do not have a strong correlation among themselves.
- The observations of yi are chosen randomly and individually from the population.

Advantages of Multiple Regression

- Multiple regression helps us to study the relationships among multiple variables in the dataset.
- The correlation between dependent and independent variables helps in predicting the outcome.
- It is one of the most convenient and popular algorithms used in machine learning.

Disadvantages of Multiple Regression

- The complexity of multiple techniques requires complex mathematical calculations.
- It is not easy to interpret the output of the multiple regression model since there are inconsistencies in the loss and error outputs.
- Multiple regression models cannot be applied to smaller datasets; they are designed for producing accurate outputs when it comes to larger datasets.

Multivariate Linear Regression

MvLR Model: Scalar Form

The multivariate (multiple) linear regression model has the form

$$y_{ik} = b_{0k} + \sum_{j=1}^{p} b_{jk} x_{ij} + e_{ik}$$

for $i \in \{1, \ldots, n\}$ and $k \in \{1, \ldots, m\}$ where

- $y_{ik} \in \mathrm{R}$ is the k-th real-valued response for the i-th observation
- $b_{0k} \in \mathrm{R}$ is the regression intercept for k-th response
- $b_{jk} \in \mathrm{R}$ is the j-th predictor's regression slope for k-th response
- $x_{ij} \in \mathrm{R}$ is the j-th predictor for the i-th observation
- $(e_{i1}, \ldots, e_{im}) \overset{\text{iid}}{\sim} \mathrm{N}(\mathbf{0}_m, \boldsymbol{\Sigma})$ is a multivariate Gaussian error vector

MvLR Model: Nomenclature

The model is multivariate because we have $m > 1$ response variables.

The model is multiple because we have $p > 1$ predictors.

- If $p = 1$, we have a multivariate simple linear regression model

The model is linear because y_{ik} is a linear function of the parameters (b_{jk} are the parameters for $j \in \{1, \ldots, p+1\}$ and $k \in \{1, \ldots, m\}$).

The model is a regression model because we are modeling response variables ($Y_1, \ldots, Y_m$) as a function of predictor variables ($X_1, \ldots, X_p$).

MvLR Model: Assumptions

The fundamental assumptions of the MLR model are:

1. Relationship between X_j and Y_k is linear (given other predictors)
2. x_{ij} and y_{ik} are observed random variables (known constants)
3. $(e_{i1}, \ldots, e_{im}) \overset{\text{iid}}{\sim} \mathrm{N}(\mathbf{0}_m, \mathbf{\Sigma})$ is an unobserved random vector
4. $\mathbf{b}_k = (b_{0k}, b_{1k}, \ldots, b_{pk})^{\mathrm{r}}$ for $k \in \{1, \ldots, m\}$ are unknown constants

Note: b_{jk} is expected increase in Y_k for 1-unit increase in X_j with all other predictor variables held constant

MvLR Model: Matrix Form

The multivariate multiple linear regression model has the form

$$\mathbf{Y} = \mathbf{X}\mathbf{B} + \mathbf{E}$$

where

- $\mathbf{Y} = [\mathbf{y}_1, \ldots, \mathbf{y}_m] \in \mathrm{R}^{n \times m}$ is the $n \times m$ response matrix
  - $\mathbf{y}_k = (y_{1k}, \ldots, y_{nk})' \in \mathrm{R}^n$ is k-th response vector $(n \times 1)$
- $\mathbf{X} = [\mathbf{1}_n, \mathbf{x}_1, \ldots, \mathbf{x}_p] \in \mathrm{R}^{n \times (p+1)}$ is the $n \times (p+1)$ design matrix
  - $\mathbf{1}_n$ is an $n \times 1$ vector of ones
  - $\mathbf{x}_j = (x_{1j}, \ldots, x_{nj})' \in \mathrm{R}^n$ is j-th predictor vector $(n \times 1)$
- $\mathbf{B} = [\mathbf{b}_1, \ldots, \mathbf{b}_m] \in \mathrm{R}^{(p+1) \times m}$ is $(p+1) \times m$ matrix of coefficients
  - $\mathbf{b}_k = (b_{0k}, b_{1k}, \ldots, b_{pk})' \in \mathrm{R}^{p+1}$ is k-th coefficient vector $(p+1 \times 1)$
- $\mathbf{E} = [\mathbf{e}_1, \ldots, \mathbf{e}_m] \in \mathrm{R}^{n \times m}$ is the $n \times m$ error matrix
  - $\mathbf{e}_k = (e_{1k}, \ldots, e_{nk})' \in \mathrm{R}^n$ is k-th error vector $(n \times 1)$

MvLR Model: Matrix Form (another look)

Matrix form writes MLR model for all *nm* points simultaneously

$$\mathbf{Y} = \mathbf{XB} + \mathbf{E}$$

$$\begin{pmatrix} y_{11} & \cdots & y_{1m} \\ y_{21} & \cdots & y_{2m} \\ y_{31} & \cdots & y_{3m} \\ \vdots & \ddots & \vdots \\ y_{n1} & \cdots & y_{nm} \end{pmatrix} = \begin{pmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1p} \\ 1 & x_{21} & x_{22} & \cdots & x_{2p} \\ 1 & x_{31} & x_{32} & \cdots & x_{3p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{np} \end{pmatrix} \begin{pmatrix} b_{01} & \cdots & b_{0m} \\ b_{11} & \cdots & b_{1m} \\ b_{21} & \cdots & b_{2m} \\ \vdots & \ddots & \vdots \\ b_{p1} & \cdots & b_{pm} \end{pmatrix} + \begin{pmatrix} e_{11} & \cdots & e_{1m} \\ e_{21} & \cdots & e_{2m} \\ e_{31} & \cdots & e_{3m} \\ \vdots & \ddots & \vdots \\ e_{n1} & \cdots & e_{nm} \end{pmatrix}$$

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Car age in years	Price in Lakhs
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12	2.2

Regression Types - Summary

Type of Regression	Description	Use Cases	Strengths	Limitations
Linear Regression	Models the relationship between a dependent variable and one/more independent variables as a straight line.	Predicting continuous variables like house prices, sales, or salaries.	Simple to implement; interpretable coefficients; works well with linear relationships.	Assumes linearity; sensitive to outliers; limited flexibility for complex relationships.
Logistic Regression	Models the probability of a binary outcome (e.g., yes/no) based on independent variables.	Binary classification tasks like spam detection or disease diagnosis.	Effective for binary outcomes; probabilistic interpretation; relatively simple to implement.	Not suitable for multi-class problems without extension.
Polynomial Regression	Extends linear regression by fitting a polynomial curve to the data.	Modeling nonlinear relationships like growth trends or trajectory prediction.	Captures complex relationships; higher flexibility than linear regression.	Prone to overfitting; sensitive to outliers; challenging to interpret coefficients.

Regression Types - Summary

Type of Regression	Description	Use Cases	Strengths	Limitations
Ridge Regression	Linear regression with L2 regularization to reduce overfitting by penalizing large coefficients.	High-dimensional data, e.g., genetics or text classification.	Reduces overfitting; balances simplicity and complexity; robust to multicollinearity.	Requires hyperparameter tuning; not suitable for sparse data.
Lasso Regression	Linear regression with L1 regularization, which performs variable selection by shrinking coefficients.	Feature selection in high-dimensional datasets like genomics or finance.	Performs automatic feature selection; reduces overfitting; interpretable.	May eliminate relevant variables; sensitive to high correlations between predictors.
Multiple Regression	Extends linear regression to include multiple independent variables.	Predicting outcomes influenced by several factors, like sales impacted by price and marketing.	Captures relationships between multiple predictors and a response variable.	Assumes linearity; requires careful variable selection; sensitive to multicollinearity.